TRAINING REGULATIONS

PRINTING SERVICES
(PREPRESS TECHNICAL OPERATIONS) NC I

SOCIAL AND OTHER COMMUNITY DEVELOPMENT SERVICES SECTOR

TECHNICAL EDUCATION AND SKILLS DEVELOPMENT AUTHORITY
East Service Road, South Luzon Expressway (SLEX), Taguig City, Metro Manila

Technical Education and Skills Development Act of 1994
(Republic Act No. 7796)

Section 22, “Establishment and Administration of the National Trade Skills Standards” of the RA 7796 known as the TESDA Act mandates TESDA to establish national occupational skill standards. The Authority shall develop and implement a certification and accreditation program in which private industry group and trade associations are accredited to conduct approved trade tests, and the local government units to promote such trade testing activities in their respective areas in accordance with the guidelines to be set by the Authority.

The Training Regulations (TR) serve as basis for the:

1 Development of curriculum and assessment instruments;
2 Registration and delivery of training programs; and
3 Competency assessment and certification

Each TR has four sections:

Section 1 **Definition of Qualification** describes the qualification and defines the competencies that comprise the qualification.

Section 2 **Competency Standards** gives the specifications of competencies required for effective work performance.

Section 3 **Training Arrangements** contains information and requirements in designing training program for certain qualification. It includes curriculum design; training delivery; trainee entry requirements; tools, equipment and materials; training facilities; trainer's qualification; and institutional assessment.

Section 4 **Assessment and Certification Arrangements** describes the policies governing assessment and certification procedures.

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PRINTING SERVICES (PREPRESS TECHNICAL OPERATIONS) NC I

SECTION 1 DEFINITION OF QUALIFICATION

The **PRINTING SERVICES (PREPRESS TECHNICAL OPERATIONS) NC I** Qualification consists of competencies that a person must achieve to perform pre-flighting procedures, perform file management, perform input processing procedures, perform line reproduction and its requirements, perform preventive maintenance of prepress equipment/machines.

The Units of Competency comprising this Qualification include the following:

Unit Code	BASIC COMPETENCIES
400311101	Receive and respond to workplace communication
400311102	Work with others
400311103	Solve/address routine problems
400311104	Enhance self-management skills
400311105	Support innovation
400311106	Access and maintain information
400311107	Follow occupational safety and health policies and procedures
400311108	Apply environmental work standards
400311109	Adopt entrepreneurial mindset in the workplace

Unit Code	COMMON COMPETENCIES
SOC532201	Update industry knowledge
SOC514202	Manage own performance
SOC514204	Maintain safe, clean and efficient work environment
ELC311203	Perform basic computer operations

Unit Code	CORE COMPETENCIES
SOC732301	Perform pre-flighting procedures
SOC732302	Perform file handling
SOC732303	Perform input processing procedures
SOC732304	Perform line reproduction and its requirements
SOC732305	Perform preventive maintenance of prepress equipment/machines

A person who has achieved this Qualification is competent to be:

- Junior Prepress Technician

SECTION 2 COMPETENCY STANDARDS

This section gives the details of the contents of the units of competency required in **PRINTING SERVICES (PREPRESS TECHNICAL OPERATIONS) NC I.**

BASIC COMPETENCIES

UNIT OF COMPETENCY	:	**RECEIVE AND RESPOND TO WORKPLACE COMMUNICATION**
UNIT CODE	:	**400311101**
UNIT DESCRIPTOR	:	This unit covers the knowledge, skills and attitudes required to receive, respond and act on verbal and written communication.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Follow routine spoken messages	1.1 Required information is gathered by listening attentively and correctly interpreting or understanding information/instructions. 1.2 Instructions/information are recorded in accordance with workplace requirements. 1.3 Instructions are acted upon immediately in accordance with information received. 1.4 Clarification is sought from workplace supervisor on all occasions when any instruction/ information is not clear.	1.1 Organizational policies/guidelines in regard to processing internal/external information 1.2 Ethical work practices in handling communications 1.3 Overview of the Communication process 1.4 Effective note-taking and questioning techniques	1.1 Conciseness in receiving and clarifying messages/ information/ communication 1.2 Accuracy in recording messages/ information 1.3 Basic communication skills 1.4 Active-listening Skills 1.5 Note-taking skills 1.6 Clarifying and probing questions (questioning skills)
2. Perform workplace duties following written notices	2.1 ***Written notices and instructions*** are read and interpreted correctly in accordance with ***organizational guidelines.***	2.1 Organizational guidelines in regard to processing internal/ external information	2.1 Conciseness in receiving and clarifying messages/ information/ communication

	2.2 Routine written instructions are followed in sequence. 2.3 Feedback is given to workplace supervisor based on the instructions/ information received.	2.2 Ethical work practices in handling communications 2.3 Overview of the Communication process 2.4 Effective questioning techniques (clarifying and probing)	2.2 Accuracy in recording messages/ information 2.3 Clarifying and probing questions (Questioning Skills) 2.4 Skills in reading and recording and labeling data 2.5 Skills in locating information

RANGE OF VARIABLES

VARIABLE	RANGE
1. Written notices and instructions	May include: 1.1 Written work instructions 1.2 Internal memos/memorandum 1.3 Business letters 1.4 External communications 1.5 Electronic mail 1.6 Briefing notes 1.7 General correspondence 1.8 Marketing materials 1.9 Guidelines/Circulars
2. Organizational guidelines	May include: 2.1 Information documentation procedures 2.2 Company guidelines and procedures 2.3 Standard Operating Procedure (SOPs) 2.4 Organization manuals 2.5 Departmental Policies and Procedures Manual 2.6 Service manual

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1 Demonstrated knowledge and understanding of organizational procedures in handling verbal and written communications 1.2 Received and acted on verbal messages and instructions correctly and efficiently 1.3 Demonstrated ability in recording instructions/information 1.4 Utilized effective clarifying and probing techniques where necessary
2. Resource Implications	The following resources should be provided: 2.1 Pens 2.2 Note pads 2.3 Computer (if applicable)
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration on communication skills (e. g., role-playing) 3.3 Oral questioning/Interview 3.3 Written Test
4. Context for Assessment	4.1 Competency may be assessed individually in the actual workplace or in a simulated environment in TESDA-accredited institutions

UNIT OF COMPETENCY : WORK WITH OTHERS

UNIT CODE : 400311102

UNIT DESCRIPTOR : This unit covers the skills, knowledge and attitudes required in working as member of a team, interacting with co-members and performing one's role in the team.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Develop effective workplace relationships	1.1 ***Duties and responsibilities*** are done in a positive manner to promote cooperation and good relationship. 1.2 Assistance is sought from ***workgroup*** when difficulties arise and addressed through discussions. 1.3 ***Feedback*** provided by others in the team is encouraged, acknowledged and acted upon. 1.4 Differences in personal values and beliefs are respected and acknowledged during interaction.	1.1 One's role, duties and responsibilities in the workplace 1.2 Acknowledging/ receiving and giving feedback 1.3 Valuing respect and empathy in the workplace 1.4 Workplace communication protocols 1.5 Teamwork 1.6 Collaboration and teambuilding within the enterprise	1.1 Communication skills – oral and written (e. g., requesting advice, receiving feedback) 1.2 Ability to relate to/interact with people from a range of social and cultural backgrounds

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
2. Contribute to work group activities	2.1 ***Support is provided to team members*** to ensure workgroup goals are met. 2.2 Constructive contributions to workgroup goals and tasks are made according to ***organizational requirements.*** 2.3 Information relevant to work is shared with team members to ensure designated goals are met.	2.1 Importance of creative collaboration, social perceptiveness and problem sensitivity in the workplace 2.2 Organizational Requirements 2.3 Importance of initiative and dedication in group process 2.4 Office and workplace technologies and automation (hardware, software systems)	2.1 Communication skills – oral and written (e. g., requesting advice, receiving feedback) 2.2 Organizing work priorities and arrangements 2.3 Team player skills 2.4 Technology skills including the ability to select and use technology appropriate to a task

RANGE OF VARIABLES

VARIABLE	RANGE
1. Duties and responsibilities	May include: 1.1 Job description and employment arrangements 1.2 Organization's policy relevant to work role 1.3 Organizational structures 1.4 Supervision and accountability requirements including OHS 1.5 Enterprise code of conduct
2. Work group	May include: 2.1 Supervisor or manager 2.2 Peers/work colleagues and clients 2.3 Other members of the organization
3. Feedback	May include: 3.1 Formal/Informal performance appraisal 3.2 Obtaining feedback from supervisors and colleagues and clients 3.3 Personal, reflective behavior strategies 3.4 Routine organizational methods for monitoring service delivery
4. Providing support to team members	May include: 4.1 Explaining/clarifying 4.2 Helping colleagues 4.3 Providing encouragement 4.4 Providing feedback to another team member 4.5 Undertaking extra tasks if necessary
5. Organizational requirements	May include: 5.1 Goals, objectives, plans, system and processes 5.2 Legal and organization policy/guidelines 5.3 OHS policies, procedures and programs 5.4 Ethical standards 5.5 Defined resources parameters 5.6 Quality and continuous improvement processes and standards

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1. Provided support to team members to ensure goals are met 1.2. Acted on feedback from clients and colleagues 1.3. Demonstrated quality/active participation in team meetings and activities
2. Resource Implications	The following resources should be provided: 2.1. Access to relevant workplace or appropriately simulated environment where assessment can take place 2.2. Materials relevant to the proposed activity or task
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Written Test 3.2 Role play 3.3 Interview/Oral questioning 3.4 Structured and unstructured activity
4. Context for Assessment	4.1. Competency assessment may occur in workplace or any appropriately simulated environment 4.2. Assessment shall be observed while task are being undertaken whether individually or in group

UNIT OF COMPETENCY : SOLVE/ADDRESS ROUTINE PROBLEMS

UNIT CODE : 400311103

UNIT DESCRIPTOR : This unit of covers the knowledge, skills and attitudes required to solve problems in the workplace including the application of problem solving techniques and to determine and resolve the root cause of routine problems.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Identify the problem	1.1 Desired operating/output parameters and expected quality of products/services are identified. 1.2 The nature of the problem by observation on routines are defined. 1.3 ***Problems*** are stated and specified clearly.	1.1 Competence includes mastery of knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations 1.2 Competence to include the ability to apply and explain fundamental causes of problems routine problems and to determine the corrective actions 1.3 Relevant equipment and operational processes 1.4 Enterprise goals, targets and measures 1.5 Enterprise quality OHS and environmental requirement 1.6 Enterprise information systems and data collation 1.7 Industry codes and standards	1.1 Using range of formal problem-solving techniques (e.g., planning, attention, simultaneous and successive processing of information) 1.2 Identifying and clarifying the nature of the problem

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
2. Assess fundamental causes of the problem	2.1 Problem-solving tool appropriate to the problem and the context is selected. 2.2 Possible causes based on experience and the use of problem-solving tools/***basic analytical techniques*** are identified. 2.3 Possible fundamental causes of problems are specified.	2.1 Competence includes a thorough knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations 2.2 Competence to include the ability to apply and explain fundamental causes of problems routine problems and to determine the corrective actions. 2.3 Relevant equipment and operational processes 2.4 Enterprise goals, targets and measures 2.5 Enterprise quality OHS and environmental requirement 2.6 Enterprise information systems and data collation 2.7 Industry codes and standards	2.1 Using range of formal problem-solving techniques (e.g., planning, attention, simultaneous and successive processing of information). 2.2 Identifying extent and causes of procedural problems.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Determine corrective action	3.1 All possible options are considered for resolution of the routine problem. 3.2 Corrective actions are determined to resolve the problem and possible future causes. 3.3 ***Action plans*** are developed identifying measurable objectives, resource needs and timelines in accordance with safety and operating procedures.	3.1 Competence includes a thorough knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations 3.2 Competence to include the ability to apply and explain, sufficient for the identification of fundamental cause, determining the corrective action and provision of recommendations 3.3 Relevant equipment and operational processes 3.4 Enterprise goals, targets and measures 3.5 Enterprise quality OHS and environmental requirement 3.6 Principles of decision making strategies and techniques 3.7 Enterprise information systems and data collation 3.8 Industry codes and standards	3.1 Using range of formal problem-solving techniques 3.2 Identifying and clarifying the nature of the problem 3.3 Devising and applying the best possible solution to a problem 3.4 Evaluating the solution
4. Communicate action plans and recommendation	4.1 Report on recommendations are prepared.	4.1 Competence includes a thorough knowledge and	4.1 Using range of formal problem solving techniques

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
s to routine problems	4.2 Recommendations are presented to ***appropriate person***. 4.3 Recommendations are followed-up, if required.	understanding of the process, normal operating parameters, and product quality to recognize non-standard situations 4.2 Competence to include the ability to apply and explain, sufficient for the identification of fundamental cause, determining the corrective action and provision of recommendations 4.3 Relevant equipment and operational processes 4.4 Enterprise goals, targets and measures 4.5 Enterprise quality, OHS and environmental requirement 4.6 Principles of decision making strategies and techniques 4.7 Enterprise information systems and data collation 4.8 Industry codes and standards	4.2 Identifying and clarifying the nature of the problem 4.3 Devising the best possible solution to a routine problem 4.4 Evaluating the solution 4.5 Developing action plans to resolving and managing routine problems

RANGE OF VARIABLES

VARIABLE	RANGE
1. Problem	May include: 1.1. Routine/non – routine processes and quality problems 1.2. Equipment selection, availability and failure 1.3. Teamwork and work allocation problem 1.4. Safety and emergency situations and incidents
2. Basic analytical techniques	May include: 2.1. Brainstorming 2.2. Case Analysis 2.3. Cause and effect diagrams 2.4. Pareto analysis 2.5. SWOT analysis 2.6. Gant chart, Pert CPM and graphs 2.7. Scattergrams
3. Action plans	May include: 3.1. Priority requirements 3.2. Measurable objectives 3.3. Resource requirements 3.4. Timelines 3.5. Co-ordination and feedback requirements 3.6. Safety requirements 3.7. Risk assessment 3.8. Environmental requirements
4. Appropriate person	May include: 4.1 Supervisor or manager 4.2 Peers/work colleagues 4.3 Other members of the organization

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1. Identified the problem. 1.2. Determined the fundamental causes of the problem. 1.3. Determined the correct / preventive action. 1.4. Developed action plans in managing routine problems. These aspects may be best assessed using project-based learning mode of assessment and case formulation.
2. Resource Implications	2.1 Assessment will require access to a workplace over an extended period, or a suitable method of gathering evidence of operating ability over a range of situations.
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1. Case Formulation 3.2. Life Narrative Inquiry (Interview) 3.3. Standardized test The unit will be assessed in a holistic manner as is practical and may be integrated with the assessment of other relevant units of competency. Assessment will occur over a range of situations, which will include disruptions to normal, smooth operation. Simulation may be required to allow for timely assessment of parts of this unit of competency. Simulation should be based on the actual workplace and will include walk through of the relevant competency components. These assessment activities should include a range of problems, including new, unusual and improbable situations that may have happened.
4. Context for Assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions

UNIT OF COMPETENCY : ENHANCE SELF-MANAGEMENT SKILLS

UNIT CODE : 400311104

UNIT DESCRIPTOR : This unit covers the knowledge, skills, and attitudes in applying the ability to regulate actions, make good decisions, and show appropriate behavior based on self-awareness.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Set personal and career goals	1.1 The difference between ***personal*** and ***career goals*** are described. 1.2 Clear and concise personal and career goals are developed. 1.3 Characteristics of motivational goals according to Locke &Latham are identified.	1.1 Definition of personal goals and career goals 1.2 SMART Model for goal setting 1.3 Five principles of goal setting according to Locke & Latham (Clarity, Challenge, Commitment, Feedback, and Task complexity)	1.1 Setting of personal and career goals 1.2 Defining, creating, and mapping personal and career goals using SMART Model for goal setting 1.3 Applying goal setting principles to personal and career goals
2. Recognize emotions	2.1 Influence that people, situations and events have on ***emotions*** are described. 2.2 Importance of responding with appropriate emotions are explained. 2.3 Influences on and consequences of emotional responses in ***social and work-related contexts*** are examined.	2.1 Common positive and negative emotions manifested in the workplace 2.2 Professional and non-professional behaviors in the workplace 2.3 Triggers and implications of positive and negative emotions in the workplace	2.1 Identifying sensitively the positive and negative emotions in the workplace 2.2 Responding with appropriate emotions in the workplace 2.3 Identifying possible consequences of inappropriate emotional responses in a social and work-related context

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Describe oneself as a learner	3.1 Factors and strategies that assist learning are identified and described. 3.2 Preferred ***learning styles*** according to VAK Learning Style Model and Kolb's Theory of Learning Styles are identified. 3.3 Range of ***learning strategies*** appropriate to specific tasks and describe work practices that assist their learning are identified and chosen.	3.1 Kolb's Theory of Learning Styles (Converger, Diverger, Assimilator, Accommodator) 3.2 VAK Learning Style Model (Visual, Auditory, Kinesthetic) 3.3 Range of learning strategies appropriate to specific tasks and describe work practices that assist their learning	3.1 Identifying factors and strategies that assist learning 3.2 Applying learning styles to positively influence school/work performance 3.3 Using appropriate learning strategies to improve study habits and learning

RANGE OF VARIABLES

VARIABLE	RANGE
1. Personal goals	May include: 1.1 Graduate from Tech-Voc training 1.2 Buy a car 1.3 Travel around the world
2. Career goals	May include but not limited to: 2.1 Graduate from Tech-Voc training 2.2 Graduate from college 2.3 Buy a car 2.4 Retire at 50 years old
3. Emotions	Positive emotions may include: 3.1 Joy 3.2 Gratitude 3.3 Hope 3.4 Serenity Negative emotions may include: 3.5 Anger 3.6 Fear 3.7 Disgust 3.8 Sadness
4. Social and work-related contexts	May include professional behavior such as: 4.1 Committed to developing and improving their skills 4.2 Professionals get the job done 4.3 They keep their word and deliver what they promise. May include non-professional behavior such as– 4.4 They engage in office politics 4.5 Bluffing and misrepresenting their skills 4.6 Blaming a colleague
5. Learning styles	May include: 5.1 Visual 5.2 Auditory 5.3 Kinesthetic 5.4 Converger 5.5 Diverger 5.6 Assimilator 5.7 Accommodator
6. Learning strategies	May include: 6.1 Explain and describe ideas with many details 6.2 Switch between ideas while studying 6.3 Use specific examples to understand abstract ideas

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1 Developed SMART personal and career goals 1.2 Applied goal setting principles 1.3 Identified sensitively the positive and negative emotions in the workplace 1.4 Responded with appropriate emotions in the workplace 1.5 Identified possible consequences of inappropriate emotional responses in a social and work-related context 1.6 Applied learning styles to positively influence school/work performance 1.7 Developed reflective practice through realization of limitations, likes/ dislikes; through showing of self-confidence
2. Resource Implications	The following resources for should be provided**:** 2.1 Access to workplace and resources
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration or simulation with oral questioning (ex. how to recognize emotions) 3.2 Case problems involving workplace diversity issues 3.3 Third-party report
4. Context for Assessment	4.1 Competency assessment may occur in workplace or any appropriately simulated environment

UNIT OF COMPETENCY : SUPPORT INNOVATION

UNIT CODE : 400311105

UNIT DESCRIPTOR : This unit of covers the knowledge, skills and attitudes required to identify, recognize and support innovation.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Identify the need for innovation in one's area of work	1.1 The value of ***innovative practices*** in the workplace is recognized. 1.2 Creativity in ***innovation*** in one's scope of work is applied. 1.3 The need for innovation in own scope of work is recognized.	1.1 Clear-cut definition of innovation 1.2 Current practice in own scope of work 1.3 Workplace procedures	1.1 Contributing in brainstorming session 1.2 Examining current practice in one's scope of work 1.3 Identifying issues and concerns of one's scope of work
2. Recognize innovative and creative ideas	2.1 Opportunities within the team are identified to develop innovation. 2.2 Creative ideas of coworkers pertaining to work practices are analyzed. 2.3 Environment conducive for learning and innovating is maintained.	2.1 Support required to generate creative ideas 2.2 Difference between innovation and creativity 2.3 Innovative climate in one's scope of work	2.1 Identifying resources required for creativity and innovation 2.2 Examining potential obstacles to and opportunities for creativity and innovation 2.3 Communicating creative ideas of co-workers
3. Support individuals' access to flexible and innovative ways of working	3.1 Individuals and key people are reinforced to identify innovative ideas to achieve outcomes. 3.2 Sharing of best practices using flexible and innovative ways of working is accomplished.	3.1 The role of employees/workers in the improvement of practices in the organization 3.2 Best practices using flexible and innovative ways of working 3.3 Obstacles in implementing innovation in one's scope of work	3.1 Encouraging co-workers to generate and develop ideas 3.2 Evaluating potential obstacles to and opportunities for creativity and innovation

	3.3 Obstacles to progress in implementing flexible and innovative ways of working are recognized.		3.3 Sharing of best practices related to innovation and creativity

RANGE OF VARIABLES

VARIABLE	RANGE
1. Innovative practices	May include: 1.1 Self-directed support 1.2 Community based services 1.3 Working within a collaborative arrangement 1.4 Making scope of work more efficient
2. Innovation	May include: 2.1 New ideas 2.2 Original ideas 2.3 Different ideas 2.4 Methods or tools

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified need for innovation in the area of work 1.2 Recognized innovative and creative ideas 1.3 Pursed agreement for flexible and innovative ways of working 1.4 Supported individuals and people to access flexible and innovative ways of working
2. Resource Implications	Specific resources for assessment 2.1. Evidence of competent performance should be obtained by observing an individual in an information management role within the workplace or operational or simulated environment.
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1. Written Test 3.2. Interview The unit will be assessed in a holistic manner as is practical and may be integrated with the assessment of other relevant units of competency. Assessment will occur over a range of situations, which will include disruptions to normal, smooth operation. Simulation may be required to allow for timely assessment of parts of this unit of competency. Simulation should be based on the actual workplace and will include walk through of the relevant competency components.
4. Context for Assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions.

UNIT OF COMPETENCY : ACCESS AND MAINTAIN INFORMATION

UNIT CODE : 400311106

UNIT DESCRIPTOR : This unit of covers the knowledge, skills and attitudes required to identify, gather, interpret and maintain information.

ELEMENT	**PERFORMANCE CRITERIA** ***Italicized terms*** are elaborated in the Range of Variables	**REQUIRED KNOWLEDGE**	**REQUIRED SKILLS**
1. Identify and gather needed information	1.1 Required ***information*** is identified based on requirements. 1.2 Sources to produce required information are identified and accessed. 1.3 Report information is collected, organized and recorded. 1.4 Organize information is collected in a way that enables easy access and retrieval by other staff.	1.1 Policies, procedures and guidelines relating to information handling in the public and private sector, including confidentiality, privacy, security, freedom of information 1.2 Data collection and management procedures 1.3 Cultural aspects of information and meaning 1.4 Sources of public sector work-related information 1.5 Public/private sector standards	1.1 Handling policies, procedures and guidelines relating to information handling in the public sector, including confidentiality, privacy, security, freedom of information 1.2 Collecting data and managing procedures 1.3 Practicing cultural aspects of information and meaning 1.4 Using public/private sector standards
2. Search for information on the internet or an intranet	2.1 Engine is search to find and select appropriate information. 2.2 Suitable techniques are used to make it easier to find useful information and to pass it on to others. 2.3 Records are use where useful information came from. 2.4 Results are used for searches of useful information.	2.1 Find and select appropriate information 2.2 Techniques in finding useful information Records are use where useful information came from 2.3 Search engines for information	2.1 Finding and selecting search engine to find and select appropriate information 2.2 Using suitable techniques to find useful information easier 2.3 Using records 2.4 Carrying out Searches

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	2.5 ***Search engine*** is chosen appropriate for the information that is needed. 2.6 Searches are carried out as per requirements.		
3. Examine information	3.1 Information and its ***sources*** are evaluated for relevance and validity to business and/or client requirements. 3.2 Information is examined as required to identify key issues. 3.3 Detailed evaluation of information is carried out as required using relevant techniques including mathematical calculations.	3.1 Data evaluation procedures 3.2 Cultural aspects of information and meaning 3.3 Sources of public sector work-related information 3.4 Public sector standards	3.1 Evaluating data 3.2 Practicing cultural aspects of information and meaning 3.3 Using public sector standards
4. Secure information	4.1 Basic file-handling techniques are used for the software 4.2 Techniques is used to handle, organize and secure information	4.1 Policies, procedures and guidelines relating to information handling in the public sector, including confidentiality, privacy, security, freedom of information 4.2 Basic file-handling techniques 4.3 Techniques in handling, organizing and saving files 4.4 Electronic and manual filing systems	4.1 Handling policies, procedures and guidelines relating to information handling in the public sector, including confidentiality, privacy, security, freedom of information 4.2 Using basic file-handling techniques is used for the software 4.3 Using different techniques in handling, organizing and saving files 4.4 Using electronic and manual filing systems

ELEMENT	**PERFORMANCE CRITERIA** ***Italicized terms*** are elaborated in the Range of Variables	**REQUIRED KNOWLEDGE**	**REQUIRED SKILLS**
5. Manage information	5.1 Information and records are maintained to ensure data and system integrity using a range of standard and complex information systems and operations. 5.2 Routine data and records are reconciled as required. 5.3 Inadequacies in system/s relating to information retrieval are identified and corrected or reported to relevant staff as required.	5.1 Policies, procedures and guidelines relating to information handling in the public sector, including confidentiality, privacy, security, freedom of information 5.2 Data collection and management procedures 5.3 Organizational information handling and storage procedures 5.4 Cultural aspects of information and meaning 5.5 Sources of public sector work-related information 5.6 Public sector standards 5.7 Databases and data storage systems	5.1 Handling policies, procedures and guidelines relating to information handling in the public sector, including confidentiality, privacy, security, freedom of information 5.2 Collecting data and managing procedures 5.3 Handling organizational information and storage procedures 5.4 Practicing cultural aspects of information and meaning 5.5 Using public sector standards 5.6 Managing databases and data storage systems

RANGE OF VARIABLES

VARIABLE	RANGE
1. Information	May include: 1.1. Property 1.2. Organizational 1.3. Technical reference
2. Search engine	May include: 2.1. Crawler-based search engine 2.1.1. Google 2.1.2. AlltheWeb 2.1.3. AltaVista 2.2. Human-powered directories 2.2.1. Yahoo directory 2.2.2. Open directory 2.2.3. Looksmart
3. Sources	May include: 3.1. Other IT systems 3.2. Manually created 3.3. Within own organization 3.4. Outside own organization 3.5. Geographically remote

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified and gathered needed information 1.2 Searched for information on the internet or an intranet 1.3 Studied and interpreted information 1.4 Handled files 1.5 Maintained information
2. Resource Implications	Specific resources for assessment 2.1. Evidence of competent performance should be obtained by observing an individual in an information management role within the workplace or operational or simulated environment.
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1. Written Test 3.2. Interview 3.3. Portfolio The unit will be assessed in a holistic manner as is practical and may be integrated with the assessment of other relevant units of competency. Assessment will occur over a range of situations, which will include disruptions to normal, smooth operation. Simulation may be required to allow for timely assessment of parts of this unit of competency. Simulation should be based on the actual workplace and will include walk through of the relevant competency components.
4. Context for Assessment	4.1. In all workplace, it may be appropriate to assess this unit concurrently with relevant teamwork or operation units.

UNIT OF COMPETENCY	**:**	**FOLLOW OCCUPATIONAL SAFETY AND HEALTH POLICIES AND PROCEDURES**
UNIT CODE	**:**	**400311107**
UNIT DESCRIPTOR	**:**	This unit covers the knowledge, skills and attitudes to identify relevant occupational safety and health policies and procedures, perform relevant occupational safety and health procedures, and comply with relevant occupational safety and health policies and standards

ELEMENT	**PERFORMANCE CRITERIA** ***Italicized terms*** are elaborated in the Range of Variables	**REQUIRED KNOWLEDGE**	**REQUIRED SKILLS**
1. Identify relevant occupational safety and health policies and procedures	1.1 Related ***occupational safety and health risks and hazards*** are recognized based on ***OSH work standards***. 1.2 ***OSH requirements/ regulations*** towards work are determined in accordance to workplace policies and procedures. 1.3 ***Incident/ Emergency procedures*** relevant to workplace are identified based on relevant OSH work standards.	1.1 Occupational safety and health risks and hazards 1.2 OSH work standards 1.3 Government approved Occupational Safety and Health Policies and regulations 1.4 Terms related to occupational safety and health 1.5 Workplace process and procedures 1.6 Standard emergency plan and procedures	1.1 Observation skills 1.2 Critical thinking skills 1.3 Communication skills

2. Perform relevant occupational safety and health procedures	2.1 Safety devices are checked in accordance with workplace OSH work standards. 2.2 ***OSH Work instructions*** are followed in accordance with workplace policies and procedures. 2.3 ***Personal protective equipment*, **materials, tools, machinery, and equipment are utilized according to OSH work standards.	2.1 OSH Work Instructions Personal protective equipment 2.2 Safe handling procedures of tools, equipment and materials 2.3 Standard emergency plan and procedures 2.4 Different OSH control measures 2.5 Standard accident and illness reporting procedures	2.1 Communication skills 2.2 Knowledge management 2.3 Organizing skills 2.4 Observation skills
3. Comply with relevant occupational safety and health policies and standards	3.1 ***Preventive Control Measures*** are identified in accordance with OSH work standards. 3.2 OSH requirements are obeyed in accordance with workplace policies and procedures. 3.3 Incident/ Emergency procedures are executed based on OSH Procedures.	3.1 OSH Preventive Control Measures 3.2 Principles of 5S 3.3 Environmental requirements relative to industrial wastes disposal 3.4 OSH requirements relative to safe handling and disposal of materials 3.5 Personal hygiene practices	3.1 Communication skills 3.2 Knowledge management 3.3 Organizing skills 3.4 Critical thinking skills 3.5 Observation skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. Occupational Safety and Health Risks and Hazards	May include: 1.1 Physical hazards – impact, illumination, pressure, noise, vibration, extreme temperature, radiation 1.2 Biological hazards- bacteria, viruses, plants, parasites, mites, molds, fungi, insects 1.3 Chemical hazards – dusts, fibers, mists, fumes, smoke, gasses, vapors 1.4 Ergonomics 1.5 Psychological factors – over exertion/ excessive force, awkward/static positions, fatigue, direct pressure, varying metabolic cycles 1.6 Physiological factors – monotony, personal relationship, work out cycle 1.7 Safety hazards (unsafe workplace condition) – confined space, excavations, falling objects, gas leaks, electrical, poor storage of materials and waste, spillage, waste and debris 1.8 Unsafe workers' act (Smoking in off-limited areas, Substance and alcohol abuse at work)
2. OSH Work Standards	May include: 2.1 OSHS Rule 1090 Hazardous Materials 2.2 OSHS Rule Gas & Electric Welding and Cutting Operations 2.3 OSHS Rule 1120 Hazardous Work Processes 2.4 OSHS Rule 1150 Materials Handling & Storage 2.5 OSHS Rule 1180 Internal Combustion Engine 2.6 OSHS Rule 1210 Electrical Safety 2.7 OSHS Rule 1420 Logging 2.8 OSHS Rule 1410 Construction Safety 2.9 OSHS Rule 1950 Pesticides & Fertilizers
3. OSH Requirements/ Regulations	May include: 3.1 Clean Air Act 3.2 Building code 3.3 National Electrical and Fire Safety Codes 3.4 Waste management statutes and rules 3.5 Permit to Operate 3.6 Philippine Occupational Safety and Health Standards 3.7 Department Order No. 13 (Construction Safety and Health) 3.8 ECC regulations 3.9 Republic Act No. 11058 – An Strengthening Compliance with Occupational Safety and Health

4. Incident and Emergency Procedures	May include: 4.1 Chemical spills 4.2 Equipment/vehicle accidents 4.3 Explosion 4.4 Fire Drill 4.5 Gas leak 4.6 Injury to personnel 4.7 Structural collapse 4.8 Earthquake drill 4.9 Toxic and/or flammable vapors emission 4.10 Evacuation 4.11 Isolation 4.12 Basic life support/CPR 4.13 Decontamination 4.14 Calling designed emergency personnel
5. OSH Work Instructions	May include: 5.1 Worker's Participation Policies 5.2 Company Environment Safety and Health Policies 5.3 Continual OSH Improvement Instructions 5.4 Education and Training 5.5 Safety and Health Policy Statements 5.6 Mission and Vision Statements 5.7 Operating Instructions and Policies
6. Personal Protective Equipment	May include: 6.1 Arm/Hand guard, gloves 6.2 Eye protection (goggles, shield) 6.3 Hearing protection (ear muffs, ear plugs) 6.4 Hair Net/cap/bonnet 6.5 Hard hat 6.6 Face protection (mask, shield) 6.7 Apron/Gown/coverall/jump suit 6.8 Anti-static suits 6.9 High-visibility reflective vest
7. Preventive Control Measures	May include: 7.1 Eliminate the hazard (i.e., get rid of the dangerous machine 7.2 Isolate the hazard (i.e. keep the machine in a closed room and operate it remotely; barricade an unsafe area off) 7.3 Substitute the hazard with a safer alternative (i.e., replace the machine with a safer one) 7.4 Use administrative controls to reduce the risk (i.e. give trainings on how to use equipment safely; OSH-related topics, issue warning signages, rotation/shifting work schedule)

	7.5 Use engineering controls to reduce the risk (i.e. use safety guards to machine) 7.6 Use personal protective equipment 7.7 Safety, Health and Work Environment Evaluation 7.8 Periodic and/or special medical examinations of workers

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1 Recognized related occupational safety and health risks and hazards 1.2 Identified incident/emergency procedures relevant to workplace 1.3 Followed the OSH work instructions in accordance with workplace policies and procedures 1.4 Utilized personal protective equipment, materials, tools, machinery, and equipment 1.5 Obeyed OSH requirements in accordance with workplace policies and procedures 1.6 Executed incident/ emergency procedures based on OSH Procedures
2. Resource Implications	The following resources should be provided: 2.1 Facilities, materials tools and equipment necessary for the activity
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Observation/Demonstration with oral questioning 3.2 Third party report
4. Context for Assessment	4.1 Competency may be assessed in the work place or in a simulated work place setting

UNIT OF COMPETENCY : APPLY ENVIRONMENTAL WORK STANDARDS

UNIT CODE : 400311108

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitude to identify environmental work hazards, follow environment work procedures and comply with environmental requirements

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Identify environmental work hazards	1.1 Related ***environmental hazards*** are recognized based on ***environmental work standards.*** 1.2 Environmental work standards are interpreted in accordance to relevant policies. 1.3 ***Required resources*** to minimize effect of environmental hazards are prepared based on relevant environmental work standards.	1.1 Environmental Hazards 1.2 Environmental Work Standards 1.3 Required Resources 1.4 OSH Standards 1.5 Fight against poverty rights 1.6 Environmental Protection 1.7 Respect of Human Rights	1.1. Critical thinking 1.2. Problem solving 1.3. Observation Skills
2. Follow environmental work procedures	2.1 ***Environmental protection*** pre-cautionary activities are practiced based on environmental work procedures. 2.2 Work activities are executed in accordance with ***Environmental work procedures.*** 2.3 Environmental Protection Post-Activities are accomplished based on environmental work procedures.	2.1 Environmental Protection 2.2 Environmental Work Procedures 2.3 Renewable Energies	2.1 Critical thinking 2.2 Problem solving 2.3 Observation Skills
3. Comply with environmental work requirements	3.1 Required resources are utilized in accordance with workplace environmental policies. 3.2 ***Environmental hazardous and non-***	3.1 Environmental Work Procedures 3.2 Environmental Laws	3.1 Critical thinking 3.2 Problem solving 3.3 Observation Skills

	hazardous *materials* are stored in accordance with ***environmental regulations.*** 3.3 Hazardous and Non-hazardous Wastes disposed according to environmental regulations.	3.3 Environmental Hazardous and Non-Hazardous Materials	

RANGE OF VARIABLES

VARIABLE	RANGE
1. Environmental Hazards	May include: 1.1 Tobacco Smoke 1.2 Asbestos 1.3 Lead 1.4 Combustion Gases 1.5 Chemicals 1.6 Pesticides 1.7 Pollutants 1.8 Contaminated Drinking Water 1.9 Noise 1.10 Dust
2. Environmental Work Standards	May include: 2.1 Air Quality Standards 2.2 Emission Standards 2.3 ISO 14001: Environmental Management System 2.4 Environmental Statements 2.5 Environmental Quality Standards 2.6 Work Environment Measurement Standard
3. Required Resources	May include: 3.1 Electric 3.2 Water 3.3 Fuel 3.4 Telecommunications 3.5 Supplies and Materials 3.6 Trash Cans 3.7 Relevant Data Sheets 3.8 Barriers or Barricades
4. Environmental Protection	May include protection against 4.1 Overconsumption of Resources 4.2 Destruction of Ecosystems 4.3 Habitat Destructions 4.4 Extinction of Wildlife 4.5 Pollutions 4.6 Water Degradation
5. Environmental Work Procedures	May include: 5.1 Environmental pollution control measures 5.2 Oil and Fuel use 5.3 Disposal and Reuse 5.4 Herbicide applications 5.5 Breed Bird Mitigation 5.6 Tree Removal Works 5.7 Erosion Protection

	5.8 Scrub Clearance 5.9 Bankside sediment clearance
6. Environmental Hazardous and Non-Hazardous Materials	May include but not limited: 6.1 Acids 6.2 Adhesives 6.3 Aerosols 6.4 Asbestos 6.5 Batteries 6.6 Chemicals 6.7 Compact fluorescent lamps 6.8 Drugs 6.9 Dyes 6.10 E-Waste 6.11 Gasoline 6.12 Grease 6.13 Lead 6.14 Motor Oil 6.15 Solvents 6.16 Weed Killers
7. Environmental Regulations	May include: 7.1 Clean Air Act 7.2 Clean Water Act 7.3 Endangered Species Act 7.4 Resource Conservation and Recovery Act 7.5 Cave Resources and Management Act 7.6 Fisheries Code 7.7 Forestry Code 7.8 Mining Act

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1. Interpreted the Environmental Work Standards 1.2. Prepared required resources to minimize effects of environmental hazards 1.3. Practiced environmental protection pre-cautionary activities 1.4. Executed work activities 1.5. Accomplished environmental protection post-activities 1.6. Stored environmental hazardous and non-hazardous materials 1.7. Disposed hazardous and non-hazardous wastes
2. Resource Implications	The following resources should be provided: 2.1. Workplace with storage facilities 2.2. Tools, materials and equipment relevant to the tasks (ex. Cleaning tools, cleaning materials, trash bags, etc.) 2.3. PPE 2.4. Manuals and references
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1. Demonstration 3.2. Oral questioning 3.3. Written examination
4. Context for Assessment	4.1. Competency assessment may occur in workplace or any appropriately simulated environment 4.2. Assessment shall be observed while task are being undertaken whether individually or in-group

UNIT OF COMPETENCY : ADOPT ENTREPRENEURIAL MINDSET IN THE WORKPLACE

UNIT CODE : 400311109

UNIT DESCRIPTOR : This unit covers the outcomes required to support and internalize an entrepreneurial mindset and observe basic entrepreneurial practices in the workplace.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Determine entrepreneurial mindset	1.1 ***Entrepreneurial mindset*** in the workplace is determined from enterprise practices and policies. 1.2 ***Entrepreneurial mindset*** in the workplace is studied and affirmed based on current enterprise practices. 1.3 Clarification from reliable ***sources*** is sought regarding entrepreneurial mindset and corporate culture.	1.1 Workplace policies and practices relating to entrepreneurship 1.2 Elements of corporate culture 1.3 Entrepreneurial mindset 1.4 Entrepreneurial practices in the workplace 1.5 Desirable attitudes: - Patience - Willingness to learn - Attention to details	1.1 Identifying entrepreneurial mindset 1.2 Studying and affirming entrepreneurial mindset 1.3 Selecting and emulating desirable entrepreneurial practices 1.4 Communication skills
2. Identify entrepreneurial practices	2.1 Entrepreneurial practices are determined based on enterprise requirements. 2.2 Entrepreneurial practices are performed following workplace and client requirements. 2.3 Cost-effective measures are complied with reference to workplace best practices.	2.1 Quality assurance practices 2.2 Workplace and client requirements 2.3 Types of cost-effective measures 2.4 Workplace quality policy 2.5 Attitude: - Patience - Attention to details	2.1 Performing quality assurance practices 2.2 Complying quality assurance requirements 2.3 Complying to cost-effective measures 2.4 Communication skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. Entrepreneurial mindset	May include workplace thinking relating to: 1.1 Economy in the use of resources 1.2 Waste management 1.3 Quality-consciousness 1.4 Cost-consciousness 1.5 Safety- and health- consciousness
2. Quality assurance practices	May include: 2.1 Use of quality procedures manual 2.2 Quality policy 2.3 Best/Good practices 2.4 Continuous improvement program
3. Reliable sources	May include: 3.1 Supervisors 3.2 Colleagues 3.3 Clients/Partners

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Demonstrated affirmation of entrepreneurial mindset 1.2 Observed entrepreneurial practices 1.3 Complied with cost effective measures
2. Resource Implications	The following resources should be provided: 2.1 Simulated or actual workplace 2.2 Tools, materials and supplies needed to demonstrate the required tasks 2.3 References and manuals
3. Methods of Assessment	Competency in this unit may be assessed through : 3.1 Written examination 3.2 Demonstration/observation with oral questioning 3.3 Third-party report
4. Context of Assessment	4.1 Competency may be assessed in workplace or in a simulated workplace setting 4.2 Assessment shall be observed while tasks are being undertaken whether individually or in-group

COMMON COMPETENCIES

UNIT OF COMPETENCY : **UPDATE INDUSTRY KNOWLEDGE**

UNIT CODE : **SOC532201**

UNIT DESCRIPTOR : This unit of competency deals with the knowledge, skills required to access, increases and update industry knowledge.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Seek and apply information on the industry	1.1 ***Sources of information*** on the industry are correctly identified and accessed. 1.2 ***Information to assist effective work performance*** is obtained in line with job requirements. 1.3 Specific information on sector of work is accessed and updated. 1.4 Industry information is correctly applied to day-to-day work activities.	**COMMUNICATION** 1.1 Industry information sources 1.2 Overview of quality assurance in the industry 1.3 Role of individual staff members	1.1 Ready skills needed to access industry information 1.2 Basic competency skills needed to access the internet
2. Update industry knowledge	2.1 Informal and/or formal research is used to update general knowledge of the industry. 2.2 Updated knowledge is shared with clients and colleagues as appropriate and incorporated into day-to-day working activities.	**COMMUNICATION** 2.1 Role of individuals in a creative endeavor member 2.2 Sources of Industry information	2.1 Time management skills 2.2 Ready skills needed to access industry information

RANGE OF VARIABLES

VARIABLE	RANGE
1. Sources of information	May include: 1.1 Media 1.2 Reference materials 1.3 Industry associations 1.3.1. Mentors 1.3.2. Training institutions 1.3.3. Technical organizations 1.4 Industry journals 1.5 Internet 1.6 Personal observation and experience
2. Information to assist effective work performance	May include: 2.1 Different sectors of the industry and the services available in each sector 2.2 Awareness on different culture 2.3 Relationship between the industry and other industries 2.4 Industry working conditions 2.5 Legislation that affects the industry 2.5.1 Dangerous Drug Act (DDA) 2.5.2 Intellectual Property Ownership (IPO) 2.5.3 Health and safety 2.5.4 Hygiene 2.5.5 Labor work practices 2.5.6 Workers' rights and compensation 2.5.7 Viewer advisory 2.5.8 Building and other related regulations 2.5.9 Other related legislations 2.6 Guilds and associations 2.7 Industrial relations issues and major organizations 2.8 Career opportunities within the industry 2.9 Work ethics 2.10 Quality assurance

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Sought and applied information on the industry 1.2 Updated industry knowledge
2. Resource Implications	The following resources should be provided: 2.1 Tools, prepress equipment/machine and facilities appropriate to processes or activity 2.2 Supplies and materials relevant to the proposed activity 2.3 Appropriate software 2.4 Sample files
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Direct observation of activities with questioning 3.2 Oral or written questioning 3.3 Case study 3.4 Interview
4. Context of Assessment	4.1 Competency may be assessed in the actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : MANAGE OWN PERFORMANCE

UNIT CODE : SOC514202

UNIT DESCRIPTOR : This unit of competency covers the knowledge, skills and attitudes in effectively managing own workload, resources and quality work.

ELEMENT	PERFORMANCE CRITERIA *Italicized* **terms** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Plan for completion of own workload	1.1 ***Tasks*** are identified according to work instructions. 1.2 ***Work order and schedules*** are designed and organized based on timelines/deadlines. 1.3 Team coordination is applied when required in completion of workload.	**COMMUNICATION** 1.1 Teamwork 1.2 Resource management 1.3 Timelines	1.1 Planning and organizing workload and resources 1.2 Communication skills
2. Maintain quality of performance	2.1 Personal performance is monitored according to established plant standards. 2.2 Advice and guidance are obtained when necessary to achieve or maintain established standards. 2.3 Guidance from management when necessary is applied to achieve according to established standards.	**COMMUNICATION** 2.1 Indicators of appropriate performance for each area of responsibility 2.2 Steps for improving or maintaining performance	2.1 Ability to observe and record performance-related concerns and information
3. Evaluate and assess own work	3.1 Actual work output is evaluated and assessed according to work instructions. 3.2 Work productivity is evaluated according to established procedures. 3.3 Feedback is obtained from relevant persons.	**COMMUNICATION** 3.1 Project Management 3.2 Process documentation	3.1 Managing work orders 3.2 Evaluating own performance 3.3 Reporting of daily production activities

RANGE OF VARIABLES

VARIABLE	RANGE
1. Tasks	May be identified through: 1.1 Assigned task 1.2 Verbal Instructions 1.3 Policy Documents 1.4 Project brief including timelines and schedules
2. Work order and schedules	May include: 2.1 Gantt charts 2.2 Production schedule 2.3 Delivery dates

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Planned for completion of own workload 1.2 Maintained quality of performance 1.3 Evaluated and assessed own work
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant venue, equipment and materials 2.2 Assignment Instructions 2.3 Logbooks 2.4 Calendar of activities 2.5 Sample liquidation and report of expenses
3. Method of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration/observation with oral questioning
4. Context of Assessment:	4.1 Competency may be assessed in actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : **MAINTAIN SAFE, CLEAN AND EFFICIENT WORK ENVIRONMENT**

UNIT CODE : **SOC514204**

UNIT DESCRIPTOR : This unit of competency covers the knowledge, skills and attitudes needed to maintain clean, safe and efficient working environment. The unit incorporates the work safety guidelines.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWL EDGE	REQUIRED SKILLS
1. Comply with safety and health regulations	1.1 ***Safety and health regulations*** are identified and complied with. 1.2 ***Policies and procedures*** are adapted and applied.	**ENVIRONMENT** 1.1 OSHS policies and standards 1.2 Fire code	1.1 Complying with health and safety regulations 1.2 Reading and comprehension
2. Assess work area	2.1 Work areas and walkways are maintained in a safe and hazard free environment. 2.2 All routines are carried out in accordance with Occupational Safety and Health Standards (OSHS). 2.3 Waste is stored and disposed of according to OSHS.	**ENVIRONMENT** 2.1 Work Hazards Policies and Procedures 2.2 OSHS policies and procedures 2.3 Waste management	2.1 Complying with health and safety regulations
3. Check and maintain tools, equipment and resources	3.1 Tools, equipment and resources are stored according to safety regulations. 3.2 Tools, equipment and resources are checked for maintenance requirements. 3.3 Tools and equipment are referred for repair as required.	**ENVIRONMENT** 3.1 Maintenance of tools and equipment 3.2 Tools, equipment and resource maintenance requirements	3.1 Checking for maintenance requirements 3.2 Storing tools and equipment

RANGE OF VARIABLES

VARIABLE	RANGE
1. Safety and Health Regulations	May include: 1.1 Clean Air Act 1.2 National Building Code 1.3 Philippine Electrical Code 1.4 Fire Code of the Philippines 1.5 Waste management statutes and rules 1.6 Philippine Occupational Safety and Health Standards 1.7 DOLE OSH related issuances 1.8 ECC regulations
2. Policies and Procedures	May include: 2.1 Hazard Policies and Procedures 2.2 Emergency, Fire and Accident Procedures 2.3 Personal Safety Procedures 2.4 Procedures for the use of personal protective clothing and equipment 2.5 Hazard Identification 2.6 Standard Operating Procedures

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Complied with safety and health regulations 1.2 Assessed work area 1.3 Checked and maintained tools, equipment and resources
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant venue, tools, equipment and resources to perform the tasks 2.2 Required operating manual/s 2.3 Safety regulations 2.4 Relevant policies and procedures
3. Method of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration/Observation with oral questioning
4. Context of Assessment	4.1 Competency may be assessed in actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : PERFORM BASIC COMPUTER OPERATIONS

UNIT CODE : ELC311203

UNIT DESCRIPTOR : This unit covers the knowledge, skills, attitudes and values needed to identify, select, and perform basic operations and maintenance of a desktop publishing system.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Identify the basic components of a desktop publishing system	1.1 ***Hardware*** of a computer are identified according to desktop publishing system requirements. 1.2 ***Software*** are identified according to desktop publishing system requirements. 1.3 ***Other peripherals*** are identified according to desktop publishing system requirements.	**TECHNOLOGY** 1.1 Basic components of a computer system 1.2 Computer specifications for desktop color publishing system 1.3 Various operating systems and platforms	1.1 Identifying kinds of computer components 1.2 Identifying computer software 1.3 Identifying computer peripherals
2. Select the appropriate hardware, software and other peripherals needed for desktop publishing	2.1 ***Appropriate hardware*** and its ***specifications*** needed for desktop publishing are selected. 2.2 ***Appropriate software*** needed for desktop publishing is selected. 2.3 Peripherals needed for desktop publishing is selected.	**TECHNOLOGY** 2.1 Appropriate hardware for desktop publishing 2.2 Appropriate software for desktop publishing	2.1 Selecting appropriate hardware and software 2.2 Selecting appropriate peripherals for desktop publishing
3. Operate the basic components of a desktop publishing system	3.1 Files in the server is accessed via ***network.*** 3.2 Appropriate software is used according to work instructions. 3.3 Other peripherals are used properly according to manufacturer's recommendation.	**TECHNOLOGY** 3.1 Desktop taskbars and icons 3.2 Local drive, partition drive and removable drive functions 3.3 Address bar and internet browsing 3.4 Networked computer connection	3.1 Accessing files 3.2 Operating appropriate software 3.3 Written and oral communications 3.4 Following manufacturer's recommendation
4. Maintain the computer	4.1 Computer is checked regularly for any ***virus***	**TECHNOLOGY**	4.1 Checking of virus and

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
system and its external parts	***and malicious software.*** 4.2 ***External parts*** of the computer are cleaned regularly. 4.3 Computer desktop is de-cluttered in accordance with file handling procedures.	4.1 Various types of virus and its effects on the computer 4.2 Various types of malicious software and its effects on the computer 4.3 File handling system **ENVIRONMENT** 4.4 Proper computer cleaning	malicious software 4.2 Removing virus that affects the system 4.3 Recovering files affected by malicious software 4.4 Cleaning the external parts of the computer

RANGE OF VARIABLES

VARIABLE	RANGE
1. Hardware	May include: 1.1 The central processing unit (CPU) 1.2 Monitor 1.3 Keyboard and mouse 1.4 Output devices
2. Software	May include: 2.1 Operating Systems 2.2 Software application 2.3 Utility program
3. Other Peripherals	May include: 3.1 Microphone 3.2 Speaker 3.3 Headset 3.4 External drives 3.5 Scanner 3.6 Barcode/Quick Response (QR) code reader 3.7 Internet modem 3.8 Pen tool 3.9 Network hub
4. Appropriate Hardware	May include: 4.1 Extended keyboard 4.2 Required computer for workstation 4.3 High-end monitor 4.4 Postscript (PS) or Printer Command Language (PCL) driven printer
5. Specifications	May include: 5.1 Required capacity and speed of Random Access Memory (RAM) 5.2 Processor clock speed and processing capacity 5.3 Hard Drive / Solid-state Drive speed and capacity 5.4 Video card speed and capacity
6. Appropriate Software	May include: 6.1 Design and layout 6.2 Illustration and drawing 6.3 Image editing and manipulation 6.4 Productivity 6.5 Portable Document Format – For viewing/review 6.6 Scanning software – for scanning
7. Network	May include: 7.1 Personal Area Network (PAN) 7.2 Local Area Network (LAN) 7.3 Wireless Local Area Network (WLAN)

VARIABLE	RANGE
	7.4 Remote or Virtual Private Network (VPN)
8. Virus and malicious software	May include: 8.1 Trojan 8.2 Worms 8.3 Hybrids and exotic forms 8.4 Ransomware 8.5 Fileless malware 8.6 Adware 8.7 Malvertising 8.8 Spyware
9. External parts	May include: 9.1 Keyboard 9.2 Monitor 9.3 Mouse pad 9.4 CPU Case 9.5 Computer fan

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified the basic components of a desktop publishing system 1.2 Selected the appropriate hardware, software and other peripherals needed for desktop publishing 1.3 Operated the basic components of a desktop publishing system 1.4 Maintained the computer system and its external parts
1. Resource Implications	The following resources should be provided: 2.1 Tools, prepress equipment/machines and facilities appropriate to processes or activity 2.2 Supplies and materials relevant to the proposed activity 2.3 Appropriate applications 2.4 Sample files
2. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Direct observation of activities with questioning 3.2 Oral or written questioning 3.3 Case study 3.4 Interview
3. Context of Assessment	4.1 Competency may be assessed in the actual workplace or at the designated TESDA Accredited Assessment Center.

CORE COMPETENCIES

UNIT OF COMPETENCY : **PERFORM PRE-FLIGHTING PROCEDURES**

UNIT CODE : **SOC732301**

UNIT DESCRIPTOR : This unit covers the knowledge skills and attitudes required to perform pre-flight procedures and the application of commonly used tools and equipment.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Inspect client's supplied materials	1.1 Client's ***supplied materials*** are identified, labeled and counted in accordance with established procedures. 1.2 ***File source*** is identified in accordance with established procedures. 1.3 File source is checked for accessibility, compatibility and for viruses in accordance with established procedures. 1.4 Supplied ***soft file*** is classified if it is valid for output and in accordance with established procedures.	**TECHNOLOGY** 1.1 Client supplied material 1.2 Faulty materials related to work 1.3 Computer operations **MATHEMATICS** 1.4 Basic math **COMMUNICATION** 1.5 Verbal and written communication 1.6 Intellectual property 1.7 Various materials & its specification **ENVIRONMENT** 1.8 6s principles (Sort, Set orders, Shine, Standardize, Sustain, Safety)	1.1 Communication skills 1.2 Interpersonal skills 1.3 Critical thinking 1.4 Comprehension skills 1.5 Recognizing various types of materials supplied by client 1.6 Operating computer
2. Gather client's specifications and requirements	2.1 ***Job specifications*** provided by client are received for reference and use. 2.2 Clients supplied materials are checked in conformance with printing condition.	**COMMUNICATION** 2.1 Job specification 2.2 Client's requirements 2.3 Flow of communication 2.4 Task planning and preparation	2.1 Gathering client's specifications 2.2 Evaluating client's requirements 2.3 Handling / keeping supplied

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	2.3 ***Client's requirements*** are reviewed from the job order as appropriate.	2.5 Information on characteristic of materials	materials with care 2.4 Planning and preparing task
3. Prepare equipment for pre-flighting	3.1 Computer to be used is prepared to perform the preflight operation. 3.2 Appropriate ***software*** is selected in accordance with client's supplied data file. 3.3 Supplied ***fonts*** are installed in compliant with established procedures.	**TECHNOLOGY** 3.1 Computer functions 3.2 Various software and their usage 3.3 Installation of identified various types of fonts	3.1 Computer skills 3.2 Using different software related to printing 3.3 Identifying and installing various fonts
4. Perform manual checking of file	4.1 Layout size, dimension, number of pages, number of colors, orientation, and contents are inspected if it conforms to client's requirement and supplied hard copy (100% Size). 4.2 Fonts are checked for any problem or substitutions compared to the hard copy provided by the client. 4.3 Linked images are checked if it is complete, at the correct ***file format,*** appropriate resolution, ***quality*** and with the correct ***color assignment*** in accordance with established procedures. 4.4 ***Bleeds*** and margins are present and are set according to the print requirements. 4.5 Texts and line drawings are checked if ***vectored*** and in the	**TECHNOLOGY** 4.1 Various file format, image resolution and color mode 4.2 Vector and Raster awareness 4.3 Image trapping procedures **ENGINEERING** 4.4 Proper binding and finishing procedures **MATHEMATICS** 4.5 Metric system and English system measurement	4.1 Converting Metric system to English system measurement and vice versa 4.2 Checking file format, correct image resolution and color mode 4.3 Identifying areas to be trimmed 4.4 Identifying raster images areas to be vectored 4.5 Identifying areas which require image trapping and overprinting

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	correct assignment in accordance with established procedures. 4.6 ***Image trapping*** is applied to eliminate printing misregistration in accordance with established procedures. 4.7 Black texts are inspected and should be in single black color.		
5. Submit pre-flight report	5.1 Written ***pre-flight report*** is submitted to authorized personnel in accordance with established procedures. 5.2 Recommendations are presented to authorized personnel in accordance with established procedures. 5.3 Supplied materials, devices and other related materials are labeled, counted, collected and returned to relevant personnel and in accordance with established procedures.	**COMMUNICATION** 5.1 Proper filing of report 5.2 Correct presentation of report 5.3 Pre-flight reports and recommendations **ENVIRONMENT** 5.4 Materials safekeeping	5.1 Recommending and suggesting solutions 5.2 Writing reports 5.3 Performing the 5s

RANGE OF VARIABLES

VARIABLE	RANGE
1. Supplied Materials	May include: 1.1 Hard Copy 1.2 Storage Device 1.3 Written Information 1.4 Die Guides
2. File Source	May include: 2.1. Internet Downloadable Files 2.2. Mobile Device 2.3. Flash Drive 2.4. CD/DVD 2.5. Memory Cards 2.6. Computer Hard Drive
3. Soft File	May include: 3.1. Productivity Files 3.2. Desktop Publishing Files 3.3. Vectored Files 3.4. Raster Files 3.5. Encapsulated Postscript File (EPS) 3.6. Postscript File (PS) 3.7. Portable Document Format file (PDF)
4. Job Specification	May include: 4.1. Size 4.2. Dimension 4.3. Materials information 4.4. Color information 4.5. Number of pages 4.6. Printing process 4.7. Binding/finishing information
5. Client's Requirements	May include: 5.1. Quality 5.2. Quantity 5.3. Deadline
6. Software	May include: 6.1. Image Manipulation and Editing Software 6.2. Vector Graphic Editing Software 6.3. Design and Lay-out Software 6.4. Portable Document File Reader/Editor 6.5. Preflight software 6.6. Productivity Software
7. Fonts	May include: 7.1. Truetype font

VARIABLE	RANGE
	7.2. Postscript font 7.3. Open Type font
8. File Format	May include: 8.1. Joint Photographic Group (.jpeg/jpg) 8.2. Encapsulated Postscript (.eps) 8.3. Portable Document Format (.pdf) 8.4. Tagged Image File Format (.tif/tiff) 8.5. Photoshop Data (.psd) 8.6. Portable Network Graphics (.png) 8.7. Scalable Vector Graphics (.svg) 8.8. Adobe InDesign file (.indd) 8.9. Adobe Illustrator file (.ai)
9. Quality	May include: 9.1. Pixelated 9.2. High Key Image 9.3. Low Key Image 9.4. Unsharpened Image 9.5. Blurred Image 9.6. Unwanted Images
10. Color Assignment	May include: 10.1. Cyan, Magenta, Yellow, Black (CMYK) 10.2. Red, Green, Blue (RGB) 10.3. Grayscale 10.4. Spot Color 10.5. Pantone Extended Gamut
11. Bleeds	May include: 11.1. Image/Layout Extension 11.2. Final Cutting Guide 11.3. Layout Margin
12. Vectored	May include: 12.1. Lines 12.2. Path 12.3. Joints 12.4. Points
13. Image Trapping	May include: 13.1. Overprinting 13.2. Choking 13.3. Spreading 13.4. Image Registration 13.5. Knockout
14. Preflight Report	May include: 14.1. Written Report 14.2. Recommendations

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified labelled and counted client's supplied materials. 1.2 Identified file source. 1.3 Checked file source for accessibility, compatibility and for viruses. 1.4 Classified the supplied soft file if valid for output. 1.5 Received job specifications provided by client. 1.6 Checked client's supplied materials if in conformance with printing condition. 1.7 Reviewed client's requirements from the job order as appropriate. 1.8 Prepared computer to be used to perform the preflight operation. 1.9 Selected appropriate software application in accordance with client's supplied data file. 1.10 Installed supplied fonts. 1.11 Inspected layout size, dimension, number of pages, number of colors, orientation, and contents if it conforms to client's requirement and supplied hard copy (100% size). 1.12 Checked fonts for any problem or substitutions compared to the hard copy provided by the client. 1.13 Checked completeness of linked images in correct file format, appropriate resolution, quality and with the correct color assignment. 1.14 Checked presence and set the bleeds and margins. 1.15 Checked texts and line drawings if vectored and in the correct assignment. 1.16 Applied image trapping to eliminate printing misregistration. 1.17 Inspected black texts and ensured it is in single black color. 1.18 Submitted written pre-flight report to authorized personnel. 1.19 Presented recommendations to authorized personnel. 1.20 Labelled, counted, collected and returned supplied materials, devices and other related materials to relevant personnel.
2. Resource Implications	The following resources should be provided: 2.1 Workshop appropriate for the unit of competency 2.2 Tools, materials and equipment, appropriate for the unit of competency 2.3 Video clips 2.4 Case work

3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration with questioning 3.2 Observation with questioning 3.3 Written Examination 3.4 Interview
4. Context for Assessment	4.1 Competency may be assessed in the actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : PERFORM FILE HANDLING

UNIT CODE : SOC732302

UNIT DESCRIPTOR : This unit covers the knowledge, discipline, attitudes and values needed to perform file handling which include transferring, accessing, handling data, using appropriate hardware and software and the application of commonly used tools and equipment.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare task to be undertaken	1.1 ***Requirements of task*** are determined according to job specifications. 1.2 Supplied ***storage devices*** are checked for compatibility of company's input device and for viruses. 1.3 Appropriate ***hardware*** and ***software*** are selected according to task assigned and required outcome. 1.4 Task is planned to ensure that file handling ***guidelines and procedures*** are followed.	**TECHNOLOGY** 1.1 Computer operations 1.2 Various storage devices 1.3 Various computer hardware **COMMUNICATION** 1.4 Work instruction 1.5 Oral and written communication	1.1 Reading and understanding work instruction 1.2 Communication Skills 1.3 Using computer software
2. Input data into computer	2.1 Data are entered into the ***computer file server*** either by appropriate software or through drag-and-drop process. 2.2 Accuracy of ***information*** is checked and saved in accordance with prepress pre-flighting procedures. 2.3 Checked ***data*** are stored in storage media using proper folder structure according to company's archiving procedure.	**TECHNOLOGY** 2.1 Various desktop publishing software functions/ applications 2.2 Pre-flighting of data **COMMUNICATION** 2.3 Intellectual Property 2.4 Data file organization	2.1 Operating computer 2.2 Using desktop software 2.3 Applying correct pre-flighting of file 2.4 Performing internet browsing 2.5 Labeling of soft files 2.6 Organizing of files

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Retrieve data from the file server	3.1 Correct data for retrieval is selected in accordance with client's ***specifications & instructions.*** 3.2 Data are ***opened*** by selecting the correct software in accordance with established procedures. 3.3 Data are retrieved from the file server in accordance with established procedures.	**TECHNOLOGY** 3.1 Fonts handling 3.2 Concept of linking graphics **MATHEMATICS** 3.3 Data file Check	3.1 Activating needed fonts 3.2 Linking of images 3.3 Comprehension skills
4. Handle data	4.1 Procedures for ***handling data*** files are implemented in accordance with established procedures. 4.2 Out-of-date and ***unnecessary contents*** are purged according to work instructions from authorized personnel. 4.3 Saved files are ***back-up*** in accordance with established procedures.	**TECHNOLOGY** 4.1 Data files backup procedures **COMMUNICATION** 4.2 Security of data file insight 4.3 Data procedures	4.1 Following written orders 4.2 Operating various backup devices 4.3 Handling backup devices with care 4.4 Critical thinking 4.5 Managing data

RANGE OF VARIABLES

VARIABLE	RANGE
1. Requirements of Task	May include: 1.1 Archiving 1.2 Retrieval 1.3 Purging 1.4 File Handling
2. Storage Devices	May include: 2.1 Internet Downloadable Files 2.2 Mobile Device 2.3 Flash Drive 2.4 CD/DVD 2.5 Computer Hard Drive 2.6 Memory Cards
3. Hardware	May include: 3.1 Computer Set 3.2 Data Storage Device
4. Software	May include: 4.1 Design and Lay-Out 4.2 Vector Graphics 4.3 Image Manipulation and Editing Software 4.4 Portable Document Format Reader
5 Guidelines and Procedures	May include: 5.1 Proper Labeling 5.2 Proper File Location 5.3 Cataloguing of File 5.4 Indexing of Files
6 Computer File Server	May include: 6.1 Stand-alone File Server 6.2 Workstation File Server 6.3 Network Assisted Server (NAS)
7 Information	May include: 7.1 Client's Name 7.2 Job Title 7.3 Job Order Number 7.4 Date Received 7.5 Production Deadline 7.6 Technical Specifications
8 Data	May include: 8.1 Raw File 8.2 Work-On-Process File 8.3 Print Ready File 8.4 File for Archiving

VARIABLE	RANGE
9 Specifications and Instructions	May include: 9.1 Client's File 9.2 Reference File 9.3 Revised File
10 Opened	May include: 10.1 File-open Procedure 10.2 Drag-and-drop Procedure 10.3 Double-click Procedure 10.4 Right-click-open-with Procedure
11 Managing Data	May include: 11.1 Assurance of confidentiality of data files 11.2 Virus check-ups
12 Unnecessary Content	May include: 12.1 File Duplicates 12.2 Malware 12.3 Irrelevant Software 12.4 Temporary Files 12.5 Irrelevant Files
13 Back-up	May include: 13.1 Manual Back-Up 13.2 Computer Generated Back-Up 13.3 Mirroring 13.4 Cloud Back-Up

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Determined requirements of task. 1.2 Checked supplied storage devices for comparability of company's input device and for viruses. 1.3 Selected appropriate hardware and software. 1.4 Planned task to ensure that file handling guidelines and procedures are followed. 1.5 Entered data into the computer file server either by appropriate software or through drag-and-drop process. 1.6 Checked and saved accuracy of information. 1.7 Stored checked data in storage media using proper folder structure. 1.8 Selected correct data for retrieval. 1.9 Opened data by selecting the correct software. 1.10 Retrieved data from the file server. 1.11 Implemented procedures for handling data files. 1.12 Backed-up saved files.
2. Resource Implications	The following resources should be provided: 4.1 Tools, prepress equipment/machine and facilities appropriate to processes or activity 4.2 Supplies and materials relevant to the proposed activity 4.3 Appropriate Softwares 4.4 Sample Files
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Direct observation of activities with questioning 3.2 Oral or written questioning 3.3 Case Study 3.4 Interview
4. Context of Assessment	4.1 Competency may be assessed in the actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : PERFORM INPUT PROCESSING PROCEDURES

UNIT CODE : SOC732303

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes in performing input processing procedures using the appropriate hardware and software and the application of commonly used tools and equipment.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Typeset / encode data and variable data information into the computer	1.1 Machines needed for typesetting and encoding are prepared according to established procedure. 1.2 Texts are typeset into the computer using ***word processing software*** in accordance with established procedures. 1.3 ***Paragraph and text styles*** are applied in accordance with established procedures. 1.4 ***Variable data*** are encoded using appropriate application.	**TECHNOLOGY** 1.1 Basic computer operations 1.2 Basic character and paragraph style 1.3 Concept and procedures of variable data **MATHEMATICS** 1.4 Basic Math 1.5 Mensuration **COMMUNICATION** 1.6 Verbal and written communication 1.7 Intellectual property	1.1 Typing words at the average speed set by the industry with minimal errors 1.2 Encoding and coding data 1.3 Applying paragraph and text styles
2. Perform Optical Character Recognition (OCR) scanning procedures	2.1 Scanner used for Optical Character Recognition scanning is ***prepared*** according to established procedures. 2.2 ***Copy materials*** to be scanned is checked for Optical Character Recognition (OCR) technology. 2.3 ***Scanner settings*** are prepared for Optical Character Recognition (OCR) procedures. 2.4 Optical Character Recognition (OCR) ***scanning*** is performed	**TECHNOLOGY** 2.1 Selection of scanner capable of Optical Character Recognition (OCR) 2.2 Optical Character Recognition (OCR) technology 2.3 Procedures on Optical Character Recognition (OCR) scanning	2.1 Checking copy materials to be scanned 2.2 Operating device for scanning 2.3 Setting Optical Character Recognition (OCR) scanning 2.4 Handling/keepin g supplied materials with care

	in accordance with established procedure.		
3. Proofread input data	3.1 Soft copy of input data is proofread in accordance with established procedures. 3.2 Errors and omissions are corrected. 3.3 Final input data is printed on a ***desktop printer.***	**TECHNOLOGY** 3.1 Production of hardcopy **COMMUNICATION** 3.2 Procedures of proofreading 3.3 Insertion of corrections	3.1 Recognizing soft file marked errors 3.2 Identifying misspelled words 3.3 Printing of hard copy

RANGE OF VARIABLES

VARIABLE	RANGE
1. Word Processing Software	May include: 1.1 MS Word 1.2 Google Docs 1.3 Adobe InDesign 1.4 Notepad 1.5 Word Perfect 1.6 Apple Text Edit
2. Paragraph and Text Styles	May include: 2.1 Capitalization of letter 2.2 Indention 2.3 Paragraph Layout 2.4 Bold, Underline and Italics 2.5 Tab Alignment 2.6 Line Spacing 2.7 Margin 2.8 Kerning
3. Variable Data	May include: 3.1 Names and Titles 3.2 Address 3.3 Personal Information 3.4 Contact Details 3.5 Serial Numbers
4. Prepared	May include: 4.1 Check Scanner Connection 4.2 Cleaning of Scanning Table 4.3 Checking of Sensing Device
5. Copy Materials	May include: 5.1 Manuscripts 5.2 Printed Copy
6. Scanner Settings	May include: 6.1 Reduction/Enlargement 6.2 Resolutions 6.3 Batch Scanning 6.4 Fonts Recognition 6.5 Image Contrast 6.6 Bitmapped, Grayscale or Colored 6.7 Output File Format 6.8 Saved File Location
7. Scanning	May include: 7.1 Copy Checking 7.2 Scanner Copy Settings 7.3 Scanning Operations

	7.4 Saving Format
8. Desktop Printer	May include: 8.1 Laser Printer 8.2 Inkjet Printer

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared machines needed for typesetting and encoding. 1.2 Typeset text into the computer using word processing software. 1.3 Encoded variable data. 1.4 Checked copy materials to be scanned for OCR technology. 1.5 Performed OCR scanning. 1.6 Proofread soft copy of input data.
2. Resource Implications	The following resource should be provided: 2.1 Work area appropriate for the unit of competency 2.2 Industry standard hardware and software 2.3 Work instruction appropriate for the unit of competency 2.4 Sample materials to be copied
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration with oral questioning 3.2 Observation with questioning 3.3 Interview
4. Context of Assessment	4.1 Competency may be assessed in the actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : **PERFORM LINE REPRODUCTION AND ITS REQUIREMENTS**

UNIT CODE : **SOC732304**

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to perform line reproduction and its requirements and using the appropriate hardware and software and the application of commonly used tools and equipment.

ELEMENT	PERFORMANCE CRITERIA ***Italicized*** *terms are elaborated in the Range of Variables*	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare requirements for line reproduction	1.1 Check is conducted on ***reproducible copies*** for line reproduction in accordance with established procedures. 1.2 ***Image capturing device*** is prepared to digitize a copy in accordance with established procedures. 1.3 Appropriate ***settings*** are set on the image capturing device in accordance with established procedures. 1.4 Check is conducted according to the ***guidelines/ instructions*** provided by client.	**TECHNOLOGY** 1.1 Differences between vector and raster image **COMMUNICATION** 1.2 Check the original copy 1.3 Check the guidelines/instructions **ENVIRONMENT** 1.4 Materials safe keeping	1.1 Conducting check of original copy 1.2 Handling materials for safe keeping 1.3 Using appropriate equipment for line reproduction 1.4 Oral and written communication
2. Convert a line copy into a printable image	2.1 Reproducible copies are converted into a digitize form in accordance with established procedures. 2.2 Rasterized image is imported to vector graphic editor software. 2.3 Tracing procedures are performed using appropriate ***tracing tools.*** 2.4 Rasterized ***text images*** are converted	**TECHNOLOGY** 2.1 Scanner setting and operation 2.2 Software application for line reproduction 2.3 Vector software application 2.4 Different drawing tools	2.1 Executing scanning operations 2.2 Performing tracing techniques 2.3 Applying drawing skills 2.4 Applying computer operating skills

ELEMENT	PERFORMANCE CRITERIA ***Italicized*** *terms are elaborated in the Range of Variables*	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	to vector images in accordance with established procedures. 2.5 Guidelines/ instructions are applied to produce a line copy.		
3. Save a line copy in an appropriate format and print	3.1 Copy for ***archiving*** purposes is saved in accordance with established procedures. 3.2 A line copy is saved for client's approval. 3.3 A ***proof*** is printed for evaluation in accordance with established procedures.	**TECHNOLOGY** 3.1 Various file format 3.2 Various machines for proofing 3.3 Operation of various proofing machines	3.1 Choosing correct file format 3.2 Selecting appropriate printer for proofing 3.3 Operating various proofing machines

RANGE OF VARIABLES

VARIABLE	RANGE
1. Reproducible Copies	May include: 1.1 Printed or Reflective Copies 1.2 Artwork Materials 1.3 Digital Files
2. Image Capturing Device	May include: 2.1 Flatbed Scanner 2.2 Digital Camera 2.3 Mobile Camera 2.4 3D Camera 2.5 Print Screen or Screen Capture
3. Settings	May include: 3.1 Aperture 3.2 Resolution 3.3 Focus 3.4 Contrast 3.5 Lens 3.6 Speed
4. Guidelines/Instructions	May include: 4.1 Color 4.2 Dimension 4.3 Element Location 4.4 Measurement 4.5 Font Size 4.6 Font Style 4.7 Leading 4.8 Kerning
5. Tracing Tools	May include: 5.1 Computer Mouse 5.2 Track Pad 5.3 Touch Screen 5.4 Pen Tablet
6. Text Images	May include: 6.1 Flattened Text Files 6.2 Layered Text Files 6.3 Scanned Text Files
7. Archiving	May include: 7.1 Raw File 7.2 Post Script File 7.3 Portable Document Format File 7.4 Encapsulated Postscript File 7.5 Scalable Vector Graphics File
8. Proof	May include: 8.1 Laser Print Copy 8.2 Ink Jet Print Copy

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Conducted checking on reproducible copies for line reproduction. 1.2 Set appropriate settings on the image capturing device. 1.3 Conducted checking according to the guidelines and instructions by client. 1.4 Converted reproducible copies into a digitized form. 1.5 Performed tracing procedures using appropriate tracing tools. 1.6 Converted rasterized text images to vector images. 1.7 Saved copy for archiving purposes. 1.8 Saved a line copy for client's approval. 1.9 Printed a proof for evaluation.
2. Methods of Assessment	Competency in this unit may be assessed through: 2.1 Demonstration with questioning 2.2 Observation with questioning 2.3 Interviews 2.4 Written Examination
3. Resource Implications	The following resources must be provided*:* 3.1 Applicable equipment and software appropriate for the unit of competency 3.2 Printed materials to be reproduced appropriate for the unit of competency 3.3 Work area appropriate for the unit of competency 3.4 Tools, supplies, materials and equipment appropriate for the unit of competency
4. Context of Assessment	4.1 Competency may be assessed in the actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : PERFORM PREVENTIVE MAINTENANCE OF PREPRESS EQUIPMENT / MACHINES

UNIT CODE : SOC732305

UNIT DESCRIPTOR : This unit covers the knowledge, discipline, attitudes and values needed to perform preventive maintenance of prepress equipment and machines.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms are elaborated in the Range of Variables*	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Perform minor preventive maintenance of equipment / machine	1.1 ***Equipment/machines*** are identified and inspected in accordance to standard maintenance procedure. 1.2 Equipment/machines are cleaned as per manufacturer's recommendation using appropriate ***cleaning materials.*** 1.3 ***Room temperature and humidity*** are controlled according to manufacturer's recommendation.	**TECHNOLOGY** 1.1 Different types of equipment / machines 1.2 Different parts of equipment / machines **ENVIRONMENT** 1.3 Awareness on health and safety procedures 1.4 Cleaning materials and its usage 1.5 5S principles 1.6 3R principles (Reduce, Reuse, Recycle)	1.1 Performing basic machine/ equipment operations 1.2 Applying appropriate cleaning materials and minor maintenance procedures 1.3 Practicing 5s 1.4 Practicing 3Rs
2. Perform major preventive maintenance of equipment /machine	2.1 ***Processing chemicals*** are changed according to manufacturer's standard maintenance procedure. 2.2 ***Computer to plate (CTP) systems*** are cleaned according to manufacturer's maintenance procedure. 2.3 ***Wastes*** are disposed according to proper waste disposal practice. 2.4 ***Selected parts*** of equipment / machine and its ***consumables*** are replaced according	**TECHNOLOGY** 2.1 Company policy in relation to relevant technology 2.2 CTP workflow systems and its operation 2.3 Relevant equipment and operational processes 2.4 Preventive maintenance procedures established in the plant **ENVIRONMENT**	2.1 Replacing processing chemicals 2.2 Disposing wastes properly 2.3 Following oral & written instructions

ELEMENT	PERFORMANCE CRITERIA ***Italicized*** *terms are elaborated in the Range of Variables*	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	to manufacturer's standard maintenance procedure.	2.5 Proper waste disposal 2.6 Awareness on health and safety procedures	
3. Perform recording of maintenance report	3.1 ***Inspection*** of equipment /machine is recorded in accordance with company's policies and procedures. 3.2 Cleaning procedures are recorded in accordance with company's policies and procedures. 3.3 Parts to be replaced are requested for replacement in accordance with company's policies and procedures. 3.4 ***Reports*** are submitted to relevant personnel for validation in accordance with established procedure.	**COMMUNICATION** 3.1 Effective communication 3.2 Procedures in recording and reporting	3.1 Written & oral communications 3.2 Follow work instructions 3.3 Attention to details 3.4 Time conscious

RANGE OF VARIABLES

VARIABLE	RANGE
1. Equipment / Machine	May include: 1.1 Computer 1.2 Desktop Printer 1.3 Plate Recording Unit 1.4 Plate Processing Unit 1.5 Plate Pre-Registering System Device 1.6 Scanner 1.7 Digital Cutting Plotter 1.8 Viewing Light Table 1.9 Flexographic Plate Imaging System 1.10 Gravure Cylinder Imaging System 1.11 Room Air-conditioning Units
2. Cleaning Materials	May include: 2.1 Cleaning Cloth 2.2 Cleaning Solution 2.3 Sponge 2.4 Light Duty Vacuum Cleaner 2.5 Blower 2.6 Camel's Hair Brush
3. Room Temperature and Humidity	May include: 3.1 20°c to 25°c temperature 3.2 30% to 50% humidity
4. Processing Chemicals	May include: 4.1 Developer 4.2 Plate Protective Gum Solutions
5. CTP Systems	May include: 5.1 Plate Recording Unit 5.2 Plate Processing Unit 5.3 Plate Staging Unit 5.4 Plate Pre-registering System
6. Wastes	May include: 6.1 Used Developing Solutions 6.2 Plate Strips 6.3 Ink Waste 6.4 Empty Ink Cartridges 6.5 Used Water 6.6 Paper Waste
7. Selected Parts	May include: 7.1 Replenishing Hose 7.2 Viewing Table Light 7.3 Processing Rollers 7.4 Ink Cartridges
8. Consumables	May include:

VARIABLE	RANGE
	8.1 Water Filter 8.2 Air Filter 8.3 Ink Pad 8.4 Printer Ink 8.5 Plotting Blade
9. Inspection	May include: 9.1 Inspection Schedules 9.2 Inspection Checklists 9.3 Inspection Procedures
10.Reports	May include: 10.1 Routine Maintenance Report 10.2 Parts Replacement Report

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified and inspected equipment/machines. 1.2 Cleaned equipment/machines as per manufacturer's recommendation using appropriate cleaning materials. 1.3 Changed processing chemicals. 1.4 Cleaned computer to plate (CTP) systems. 1.5 Disposed wastes properly. 1.6 Recorded inspection of equipment/machines. 1.7 Recorded cleaning procedures. 1.8 Requested replacement for parts that needs to be replaced. 1.9 Submitted reports to relevant personnel for validation.
2. Methods of Assessment	Competency in this unit may be assessed through: 2.1 Demonstration with questioning 2.2 Interviews 2.3 Written examination
3. Resource Implications	The following resources must be provided: 3.1 Tools, prepress equipment/machine and facilities appropriate to processes or activity 3.2 Supplies and materials relevant to the unit of competency 3.3 Workplace or Work Area relevant to the unit of competency
4. Context of Assessment	4.1 Competency may be assessed in the actual workplace or at the designated TESDA Accredited Assessment Center.

SECTION 3 TRAINING ARRANGEMENTS

These standards are set to provide technical and vocational education and training (TVET) providers with information and other important requirements to consider when designing training programs for **PRINTING SERVICES (PREPRESS TECHNICAL OPERATIONS) NC I.**

3.1 CURRICULUM DESIGN

TESDA shall provide the training on the development of competency-based curricula to enable training providers develop their own curricula with the components mentioned below.

Delivery of knowledge requirements for the basic, common and core units of competency specifically in the areas of mathematics, science/technology, communication/language and other academic subjects shall be contextualized. To this end, TVET providers shall develop a Contextual Learning Matrix (CLM) to accompany the curricula.

Course Title: PRINTING SERVICES (PREPRESS TECHNICAL OPERATIONS) **NC Level: NC I**

Nominal Training Duration:

47	hours	Basic Competencies
45	hours	Common Competencies
308	hours	Core Competencies
400	**hours**	**TOTAL**

Course Description:

This course is designed to provide the learner with knowledge, skills and attitude, applicable in performing work activities involve in performing pre-flighting procedures, performing file management, performing input processing procedures, performing line reproduction and its requirements, performing preventive maintenance of prepress equipment/machines

Upon completion of the course, the learners are expected to demonstrate the above-mentioned competencies to be employed. To obtain this, all units prescribed for this qualification must be achieved.

BASIC COMPETENCIES
(47 HOURS)

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
1. Receive and respond to workplace communication	1.1 Follow routine spoken messages	• Exercise Conciseness in receiving and clarifying messages/ information/ communication	• Group discussion • Interaction • Reportorial • Modular	• Interviews/ • Questioning • Practical/ • Performance Test • Observation	4 hours
	1.2 Perform workplace duties following written notices	• Practice Accuracy in following written/ oral instruction/ information • Practice written and oral communication skills • Case Study in handling written communication • Practice relaying/ disseminating messages/ information • Analyze different messages	• Lecture/ • Discussion • Demonstration • Case Study	• Written • Practical • Written • Demonstration	4 hours
2. Work with others	2.1 Develop effective workplace relationships	• Read job description and organizations policies relevant to work role • Read personnel code of conduct and discipline • Role play on cooperation and good relationship • Study table of organization and identify team members • Role play on team work. • Role play on receiving feedback from supervisor • Role play on providing feedback. • Listen to lecture on Valuing and exemplifying respect and empathy in the workplace	• Individual Work • Discussion • Role Play • Lecture	• Role Play • Structured activity • Written Test	2 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	2.2 Contribute to work group activities	• Discussion on creative collaboration, social perceptiveness and problem sensitivity • Role play on creative collaboration, social perceptiveness and problem sensitivity. • Participate in a goal setting activity • Participate in planning and implementation of a group activity. • Participate in evaluation of the group activity	• Lecture/ Discussion • Role Play • Group Work	• Role Play • Structured activity • Written Test	1 hour
3. Solve/address routine problems	3.1 Identify the problem	• Show mastery of the current industry hardware and software products and services - Show mastery of knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations - Relevant equipment and operational processes - Enterprise goals, targets and measures - Enterprise quality OHS and environmental requirement - Enterprise information systems and data collation - Industry codes and standards • Use range of formal problem-solving techniques (e.g., planning, attention, simultaneous and successive processing of information) • Identify and clarify the nature of the problem	• Interactive Lecture • Appreciative Inquiry • Demonstration	• Case Formulation • Life Narrative Inquiry (Interview) • Standardized test	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	3.2 Assess fundamental causes of problem	• Show mastery of the current industry hardware and software products and services - Show mastery of knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations - Relevant equipment and operational processes - Enterprise goals, targets and measures - Enterprise quality OHS and environmental requirement - Enterprise information systems and data collation - Industry codes and standards • Use range of formal problem-solving techniques (e.g., planning, attention, simultaneous and successive processing of information) • Identify and clarify the nature of the problem	• Group discussion • Lecture • Demonstration • Role play	• Case Formulation • Life Narrative Inquiry (Interview) • Standardized test	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	3.3 Determine corrective action	• Show mastery of the current industry hardware and software products and services - Show mastery of knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations - Relevant equipment and operational processes - Enterprise goals, targets and measures - Enterprise quality OHS and environmental requirement - Enterprise information systems and data collation - Industry codes and standards • Use range of formal problem-solving techniques (e.g., planning, attention, simultaneous and successive processing of information) • Identify and clarify the nature of the problem	• Group discussion • Lecture • Demonstration • Role play	• Case Formulation • Life Narrative Inquiry (Interview) • Standardized test	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	3.4 Communicate action plans and recommendations to routine problems	• Show mastery of the current industry hardware and software products and services - Show mastery of knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations - Relevant equipment and operational processes - Enterprise goals, targets and measures - Enterprise quality OHS and environmental requirement - Enterprise information systems and data collation - Industry codes and standards • Use range of formal problem-solving techniques (e.g., planning, attention, simultaneous and successive processing of information) • Identify and clarify the nature of the problem	• Group discussion • Lecture • Demonstration • Role playing	• Case Formulation • Life Narrative Inquiry (Interview) • Standardized test	1 hour
4. Enhance Self-Management Skills	4.1 Set personal and career goals	• Define and set personal goals and career goals • Describe the SMART Model for goal setting • Create personal and career goals using SMART Model for goal setting • Explain and apply the principles of goal setting according to Locke & Latham	• Discussion • Making of personal and career goals by students • Brainstorming	• Demonstration or simulation with oral questioning • Case problems involving workplace diversity issues	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	4.2 Recognize emotions	• Identify common positive and negative emotions manifested in the workplace • Distinguish professional and non-professional behaviors in the workplace • Recognize triggers and implications of positive and negative emotions in the workplace • Respond with appropriate emotions and identify possible consequences of inappropriate emotional responses in a social and work-related context	• Discussion • Interactive Lecture • Brainstorming	• Demonstration or simulation with oral questioning • Case problems involving workplace diversity issues	1 hour
	4.3 Describe oneself as a learner	• Review Kolb's Theory of Learning Styles • Describe VAK Learning Style Model (Visual, Auditory, Kinesthetic) • Cite learning strategies appropriate to specific tasks and describe work practices that assist learning • Identify factors and strategies that assist learning • Apply learning styles to positively influence school/work performance • Use appropriate learning strategies to improve study habits and learning	• Discussion • Interactive Lecture • Brainstorming • Simulation	• Demonstration or simulation with oral questioning • Case problems involving workplace diversity issues	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
5. Support Innovation	5.1 Identify the need for innovation in one's area of work	• Show mastery of the clear-cut definition of innovation and its characteristics • Identify the need for innovation in one's work area • Identify work procedures needing change • Contribute to brainstorming sessions with co-workers on identifying tasks needing change	• Interactive Lecture • Appreciative Inquiry • Demonstration • Group work	• Psychological and behavioral Interviews • Performance Evaluation • Life Narrative Inquiry • Review of portfolios of evidence and third-party workplace reports of on-the-job performance. • Standardized assessment of character strengths and virtues applied	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	5.2 Recognize innovative and creative ideas	• Identify resources needed for change and potential obstacles as well • Show positive attitudes and behaviors in accepting and in needing change in one's work area • Delineate differences between creativity and innovation	• Interactive Lecture • Appreciative Inquiry • Demonstration • Group work	• Psychological and behavioral Interviews • Performance Evaluation • Life Narrative Inquiry • Review of portfolios of evidence and third-party workplace reports of on-the-job performance. • Standardized assessment of character strengths and virtues applied	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	5.3 Support individuals' access to flexible and innovative ways of working	• Identify different roles of employees/workers in the improvement of practices in the organization • Identify practices for flexible and innovative ways of working • Share information with co-workers • Detect potential problems in implementing flexible ways of working	• Interactive Lecture • Appreciative Inquiry • Demonstration • Group work	• Psychological and behavioral Interviews • Performance Evaluation • Life Narrative Inquiry • Review of portfolios of evidence and third-party workplace reports of on-the-job performance. • Standardized assessment of character strengths and virtues applied • Review of portfolios of evidence and third-party workplace reports of on-the-job performance. • Standardized assessment of character strengths and virtues applied	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
6. Access and maintain information	6.1 Identify and gather needed information	• Lecture and discussion on: - Policies, procedures and guidelines relating to information handling in the public and private sector, including confidentiality, privacy, security, freedom of information - Data collection and management procedures - Public/private sector standards • Identify sources to produce required information • Perform exercises on information gathering	• Lecture • Demonstration • Practical exercises	• Oral evaluation • Written Test • Observation	3 hours
	6.2 Search for information on the internet or an intranet	• Lecture and discussion on: - Techniques in finding useful information - Search engines for information • Find and select appropriate information • Perform information searching on the internet using different search engines	• Group discussion • Lecture • Demonstration • Practical exercises	• Oral evaluation • Written Test • Observation • Presentation	2 hours
	6.3 Examine information	• Lecture and discussion on: - Data evaluation procedures - Cultural aspects of information and meaning - Sources of public sector work-related information • Evaluation of searched information	• Group discussion • Lecture • Demonstration • Practical exercises	• Oral evaluation • Written Test • Observation • Presentation	2 hours
	6.4 Secure information	• Lecture and discussion on: - Basic file-handling techniques - Techniques in handling, organizing and saving files - Electronic and manual filing systems • Performance of basic file-handling techniques • Application of electronic and manual filing systems	• Group discussion • Lecture • Demonstration • Role Play • Practical exercises	• Oral evaluation • Written Test • Observation • Presentation	3 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	6.5 Manage information	• Lecture and discussion on: - Organizational information handling and storage procedures - Databases and data storage systems • Managing databases and data storage systems	• Group discussion • Lecture • Demonstration • Practical exercises	• Oral evaluation • Written Test • Observation • Presentation	2 hours
7. Follow Occupational Safety and Health Policies and Procedures	7.1 Identify relevant occupational safety and health policies and procedures	• Discussion of Risks and Hazards • Risk and Hazard Identification	• Lecture • Group Discussion	• Written Exam • Demonstration • Observation • Interviews / • Questioning	2 hours
	7.2 Perform relevant occupational safety and health procedures	• Demonstration of proper use of Personal Protective Equipment and Materials Handling • Practice Emergency Plan	• Lecture • Group Discussion	• Written Exam • Demonstration • Observation • Interviews / • Questioning	2 hours
	7.3 Comply with relevant occupational safety and health policies and standards	• Discussion on Personal Hygiene and Preventive Control Measures • Practice 5S and waste segregation	• Lecture • Group Discussion	• Written Exam • Demonstration • Observation • Interviews / • Questioning	4 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
8. Apply Environmental Work Standards	8.1 Identify environmental work hazards	• Discussions in - Reduction in greenhouse gas emissions, - Increase the share of renewables of gross final energy consumption, - Long-term reduction of energy consumption, - Release of materials into the environment should, in the long run, not exceed the adaptability of the eco-system, - Dangers and unjustifiable risks to human health • Energy and natural resource consumption and the provision of transport services	• Lecture • Group Discussion	• Written Exam • Demonstration • Observation • Interviews / • Questioning	1 hour
	8.2 Follow environmental work procedures	• Discussions Protection against - Human Dangers - Overconsumption of Resources - Destruction of Ecosystems - Habitat Destructions - Extinction of Wildlife - Pollutions • Water Degradation	• Lecture • Group Discussion • Demonstration	• Written Exam • Demonstration • Observation • Interviews / • Questioning	1 hour
	8.3 Comply with environmental work requirements	• Discussions Environmental Regulations and its requirements relevant to the industry and work activities • Demonstration and Practice Environmental Compliance	• Lecture • Group Discussion • Demonstration	• Written Exam • Demonstration • Observation • Interviews / Questioning	1 hour
9. Adopt Entrepreneurial Mindset in the Workplace	9.1 Determine entrepreneurial mindset	• Discussion on Entrepreneurial Mindset • Games to develop entrepreneurial mind set	• Lecture discussion • Games	• Written Test • Role play	2 hours
	9.2 Identify entrepreneurial practices	• Case study- quality assurance practices • Discussion on cost effective measures • Discussion on Workplace quality Policy	• Case study • Lecture discussion	• Written Test • Case Study	1 hour

COMMON COMPETENCIES
(45 HOURS)

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
1. Update industry knowledge (11 hrs)	1.1 Seek and apply information on the industry	• Lecture and discussion on the following topics: ○ Industry information sources ○ Overview of quality assurance in the industry ○ Role of individual staff members • Perform the following tasks: ○ Identify and access sources of information on the industry ○ Obtain information to assist effective work performance in line with job requirements ○ Access and update specific information on sector of work ○ Correctly apply Industry information to day-to-day work activities ○ Ready skills needed to access industry information ○ Basic competency skills needed to access the internet	• Lecture-discussion • Demonstration • Hands-on/ Writeshop • Seminars/ conferences	• Written test • Oral Questioning • Presentation • Evaluation of written output	8 hours
	1.2 Update industry knowledge	• Lecture and discussion on the following topics: ○ Role of individuals in a creative endeavor member ○ Sources of Industry information • Perform the following tasks: ○ Use informal and/or formal research to update general knowledge of the industry	• Lecture-discussion • Demonstration • Hands-on/ Writeshop • Research	• Written test • Oral Questioning • Presentation • Evaluation of written output	3 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Share updated knowledge with clients and colleagues as appropriate and incorporate into day-to- day working activities ○ Time management skills ○ Ready skills needed to access industry information			
2. Manage own performance (8 hrs)	2.1 Plan for completion of own workload	• Lecture and discussion on the following topics: ○ Teamwork ○ Resource management ○ Timelines • Perform the following tasks: ○ Identify tasks are identified according to work instructions ○ Design and organize work order and schedules based on timelines/deadlines ○ Apply team coordination when required in completion of workload ○ Planning and organizing workload and resources ○ Communication skills	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	3 hours
	2.2 Maintain quality of performance	• Lecture and discussion on the following topics: ○ Indicators of appropriate performance for each area of responsibility ○ Steps for improving or maintaining performance • Perform the following tasks: ○ Monitor personal performance	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	2 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		according to established plant standards ○ Obtain advice and guidance when necessary to achieve or maintain established standards ○ Apply guidance from management when necessary to achieve according to established standards			
	2.3 Check and assess own work	• Lecture and discussion on the following topics: ○ Project Management ○ Process documentation • Perform the following tasks: ○ Check and assess actual work output according to work instructions ○ Check work productivity according to established procedures ○ Obtain feedback from relevant persons ○ Handling work orders ○ Checking own performance ○ Reporting of daily production activities	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	3 hours
3. Maintain safe, clean, and efficient work environment (13 hrs)	3.1 Comply with safety and health regulations	• Lecture and discussion on the following topics: ○ OSHS policies and standards ○ Fire code • Perform the following tasks: ○ Identify and comply with safety and health regulations ○ Adapt and apply policies and procedures	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	6 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		o Complying with health and safety regulations o Reading and comprehension			
	3.2 Assess work area	• Lecture and discussion on the following topics: o Work Hazards Policies and Procedures o OSHS policies and procedures o Waste management • Perform the following tasks: o Maintain work areas and walkways in a safe and hazard free environment o Carry out all routines in accordance with Occupational Safety and Health Standards (OSHS) o Store and dispose according to OSHS o Complying with health and safety regulations	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	5 hours
	3.3 Check and maintain tools, equipment and resources	• Lecture and discussion on the following topics: o Maintenance of tools and equipment o Tools, equipment and resource maintenance requirements • Perform the following tasks: o Store tools, equipment and resources according to safety regulations o Check tools, equipment and resources for maintenance requirements o Refer tools and equipment for repair as required o Checking for maintenance	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	2 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		requirements ○ Storing tools and equipment			
4. Perform basic computer operations (13 hrs)	4.1 Identify the basic components of a desktop publishing system	• Lecture and discussion on the following topics: ○ Basic components of a computer system ○ Computer specifications for desktop color publishing system ○ Various operating systems and platforms • Perform the following tasks: ○ Identify hardware of a computer according to desktop publishing system requirements ○ Identify software according to desktop publishing system requirements ○ Identify other peripherals according to desktop publishing system requirements ○ Identifying kinds of computer components ○ Identifying computer software ○ Identifying computer peripherals	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	5 hours
	4.2 Select the appropriate hardware, software and other peripherals needed for desktop publishing	• Lecture and discussion on the following topics: ○ Appropriate hardware for desktop publishing ○ Appropriate software for desktop publishing • Perform the following tasks: ○ Select appropriate hardware and its specifications needed for desktop publishing	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	3 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Select appropriate software needed for desktop publishing ○ Select peripherals needed for desktop publishing ○ Selecting appropriate hardware and software ○ Selecting appropriate peripherals for desktop publishing			
	4.3 Operate the basic components of a desktop publishing system	• Lecture and discussion on the following topics: ○ Desktop taskbars and icons ○ Local drive, partition drive and removable drive functions ○ Address bar and internet browsing ○ Networked computer connection • Perform the following tasks: ○ Access files in the server via network ○ Use appropriate software according to work instructions ○ Use properly other peripherals according to manufacturer's recommendation ○ Accessing files ○ Operating appropriate software ○ Written and oral communications ○ Following manufacturer's recommendation	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	3 hours
	4.4 Maintain the computer system and its external parts	• Lecture and discussion on the following topics: ○ Various types of virus and its effects on the computer ○ Various types of malicious software and its effects on the computer ○ File handling system	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	2 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Proper computer cleaning • Perform the following tasks: ○ Check computer regularly for any virus and malicious software ○ Clean external parts of the computer regularly ○ De-clutter computer desktop in accordance with file management procedures ○ Checking of virus and malicious software ○ Removing virus that affects the system ○ Recovering files affected by malicious software ○ Cleaning the external parts of the computer			

CORE COMPETENCIES
(308 HOURS)

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
1. Perform pre-flighting procedures (112 hours)	1.1 Inspect client's supplied materials	• Lecture and discussion on the following topics: ○ Client supplied material ○ Faulty materials related to work ○ Computer operations ○ Basic math ○ Verbal and written communication ○ Intellectual property ○ Various materials and its specification ○ 6s principles (Sort, Set orders, Shine, Standardize, Sustain, Safety) • Perform the following tasks: ○ Identify, label and count client's supplied materials in accordance with established procedures ○ Identify file source in accordance with established procedures ○ Handle file source for accessibility, compatibility and for viruses in accordance with established procedures ○ Classify supplied soft file if it is valid for output and in accordance with established procedures ○ Demonstrate communication skills ○ Demonstrate interpersonal skills ○ Demonstrate critical thinking ○ Demonstrate analytical and comprehension skills ○ Recognize various types of materials supplied by client	• Lecture • Group discussion • Demonstration • Case work	• Oral questioning/ interview • Written Test • Candidate's demonstration • Direct observation of candidate • Hands on	12 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Operating computer			
	1.2 Gather client's specifications and requirements	• Lecture and discussion on the following topics: ○ Job specification ○ Client requirements ○ Flow of communication ○ Task planning and preparation ○ Information on characteristic of materials • Perform the following tasks: ○ Receive job specifications provided by client for reference and use ○ Check clients supplied materials in conformance with printing condition ○ Check client's requirements from the job order as appropriate ○ Gather client's specifications ○ Check client's requirements ○ Handle / keep supplied materials with care ○ Plan and prepare task	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate • Hands on	8 hours
	1.3 Prepare equipment for pre-flighting	• Lecture and discussion on the following topics: ○ Computer functions ○ Various software and their usage ○ Installation of identified various types of fonts • Perform the following tasks: ○ Prepare computer to be used to perform the preflight operation ○ Select correct software in accordance with client's supplied data file	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Return demonstration • Direct observation of candidate • Hands on	36 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Install supplied fonts in compliant with established procedures ○ Demonstrate computer skills ○ Use different software related to printing ○ Identify and install various fonts			
	1.4 Perform manual checking of file	• Lecture and discussion on the following topics: ○ Various file format, image resolution and color mode ○ Vector and Raster awareness ○ Image trapping procedures ○ Proper binding and finishing procedures ○ Metric system and English system measurement • Perform the following tasks: ○ Inspect layout size, dimension, number of pages, orientation, and contents if it conforms to client's requirement and supplied hard copy (100% Size) ○ Check fonts for any problem or substitutions compared to the hard copy provided by the client ○ Check linked images if it is complete, at the correct file format, appropriate resolution, quality and with the correct color assignment in accordance with established procedures ○ Set bleeds and margins if present according to the print requirements ○ Check texts and line drawings if vectored and in the correct assignment in accordance with established procedures	• Case work • Project method • Lecture • Group discussion • Simulation • Role playing • Video presentation	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate • Hands on	48 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Apply image trapping to eliminate printing misregistration in accordance with established procedures ○ Inspect black texts if in single black color ○ Convert Metric system to English system measurement and vice versa ○ Check file format, correct image resolution and color mode ○ Identify areas to be trimmed ○ Identify raster images areas to be vectored ○ Identify areas which require image trapping and overprinting			
	1.5 Submit pre-flight report	• Lecture and discussion on the following topics: ○ Proper filing of report ○ Correct presentation of report ○ Pre-flight reports and recommendation ○ Materials safekeeping • Perform the following tasks: ○ Submit written pre-flight report to authorized personnel in accordance with established procedures ○ Present recommendations to authorized personnel in accordance with established procedures ○ Label, count, collect and return supplied materials, devices and other related materials to relevant personnel and in accordance with established procedures ○ Recommend and suggest solutions ○ Demonstrate writing reports ○ Perform the 5s	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Return demonstration • Direct observation of candidate • Hands on	8 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
2. Perform file handling (40 hours)	2.1 Prepare task to be undertaken	• Lecture and discussion on the following topics: ○ Computer operations ○ Various storage devices ○ Various computer hardware ○ Work instruction ○ Oral and written communication • Perform the following tasks: ○ Determine requirements of task according to job specifications ○ Check supplied storage devices for compatibility of company's input device and for viruses ○ Select appropriate hardware and software according to task assigned and required outcome ○ Plan task to ensure that file handling guidelines and procedures are followed ○ Read and understand work instruction ○ Demonstrate communication skills ○ Use computer software	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate	8 hours
	2.2 Input data into computer	• Lecture and discussion on the following topics: ○ Various desktop publishing software functions/ applications ○ Pre-flighting of data ○ Intellectual Property ○ Data file organization • Perform the following tasks: ○ Enter data into the computer file server either by appropriate software or through drag-and-drop process	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate	8 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Check and save accuracy of information in accordance with prepress pre-flighting procedures ○ Store checked data in storage media using proper folder structure according to company's archiving procedure ○ Operate computer ○ Use desktop software ○ Apply correct pre-flighting of file ○ Perform internet browsing ○ Label of soft files ○ Organize of files			
	2.3 Retrieve data from the file server	• Lecture and discussion on the following topics: ○ Fonts handling ○ Concept of linking graphics ○ Data file check • Perform the following tasks: ○ Select correct data for retrieval in accordance with client's specifications & instructions ○ Open data by selecting the correct software in accordance with established procedures ○ Retrieve data from the file server in accordance with established procedures ○ Activate needed fonts ○ Link of images	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate	16 hours
	2.4 Handle data	• Lecture and discussion on the following topics: ○ Data files backup procedure ○ Security of data file insight ○ Data handling procedures	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration	8 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		• Perform the following tasks: ○ Implement procedures for managing data files in accordance with established procedures ○ Purge out-of-date and unnecessary contents according to work instructions from authorized personnel ○ Back-up saved files in accordance with established procedures ○ Follow written orders ○ Operate various backup devices ○ Handle backup devices with care ○ Demonstrate critical thinking ○ Handle data		• Direct observation of candidate	
3. Perform input processing procedure (64 hours)	3.1 Typeset / encode data and variable data information into the computer	• Lecture and discussion on the following topics: ○ Basic computer operations ○ Basic character and paragraph style ○ Concept and procedures of variable data ○ Basic Math ○ Mensuration ○ Verbal and written communication ○ Intellectual property • Perform the following tasks: ○ Prepare machines needed for typesetting and encoding according to established procedure ○ Typeset texts into the computer using word processing software in accordance with established procedures ○ Apply paragraph and text styles in accordance with established procedures	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate • Hands on	24 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Encode variable data using appropriate software ○ Type/encode words at the average speed set by the industry with minimal errors ○ Encode and code data ○ Apply paragraph and text styles			
	3.2 Perform Optical Character Recognition (OCR) scanning procedures	• Lecture and discussion on the following topics: ○ Selection of scanner capable of Optical Character Recognition (OCR) ○ Optical Character Recognition (OCR) technology ○ Procedures on Optical Character Recognition (OCR) scanning • Perform the following tasks: ○ Prepare scanner used for Optical Character Recognition scanning according to established procedures ○ Check copy materials for Optical Character Recognition (OCR) technology scanning ○ Prepare scanner settings for Optical Character Recognition (OCR) procedure ○ Perform Optical Character Recognition (OCR) scanning procedure in accordance with established procedure ○ Check copy materials to be scanned ○ Operate device for scanning ○ Set Optical Character Recognition (OCR) scanning ○ Handle/keep supplied materials with care	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate • Hands on	24 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	3.3 Proofread input data	• Lecture and discussion on the following topics: ○ Production of hardcopy ○ Procedures of proofreading ○ Insertion of corrections • Perform the following tasks: ○ Proofread soft copy of input data in accordance with established procedures ○ Correct errors and omissions ○ Print final input data on a desktop printer ○ Recognize soft file marked errors ○ Identify misspelled words ○ Print of hard copy	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate • Hands on	16 hours
4. Perform line reproduction and its requirements (64 hours)	4.1 Prepare requirements for line reproduction	• Lecture and discussion on the following topics: ○ Differences between vector and raster image ○ Check on the original copy ○ Check on the guidelines/instructions • Perform the following tasks: ○ Conduct checking on reproducible copies for line reproduction in accordance with established procedures ○ Prepare image capturing device to digitize a copy in accordance with established procedures ○ Set appropriate settings on the image capturing device in accordance with established procedures ○ Conduct checking according to the guidelines/instructions provided by client ○ Conduct checking of original copy ○ Handle materials for safe keeping.	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate	12 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Use appropriate equipment for line reproduction ○ Demonstrate oral and written communication			
	4.2 Convert a line copy into a printable image	• Lecture and discussion on the following topics: ○ Scanner setting and operation ○ Software application for line reproduction ○ Vector software application ○ Different drawing tools • Perform the following tasks: ○ Convert reproducible copies into a digitize form in accordance with established procedures ○ Import rasterized image to Adobe Illustrator ○ Perform tracing procedures using appropriate tracing tools ○ Convert rasterized text images to vector image in accordance with established procedures ○ Apply guidelines/instructions to produce a line copy ○ Perform scanning operations ○ Perform tracing techniques ○ Apply drawing skills ○ Apply computer operating skills	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate • Hands on	48 hours
	4.3 Save a line copy in an appropriate format and print	• Lecture and discussion on the following topics: ○ Various file format ○ Various machines for proofing ○ Operation of various proofing machine • Perform the following tasks:	• Case work • Lecture	• Oral questioning/ interview • Written test • Return demonstration	4 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Save a copy for archiving purposes in accordance with established procedures ○ Save a line copy for client's approval ○ Print a proof for evaluation in accordance with established procedures ○ Choose correct file format ○ Select appropriate printer for proofing ○ Operate various proofing machine		• Direct observation of candidate • Hands on	
5. Perform preventive maintenance of prepress equipment / machines (28 hours)	5.1 Perform minor preventive maintenance of equipment / machine	• Lecture and discussion on the following topics: ○ Different types of equipment / machine ○ Different parts of equipment / machine ○ Awareness on health and safety procedures ○ Cleaning materials and its usage ○ 5S principles ○ 3R principles (Reduce, Reuse, Recycle) • Perform the following tasks: ○ Identify and inspect equipment/machines in accordance to standard maintenance procedure ○ Clean equipment/machines as per manufacturer's recommendation using appropriate cleaning materials ○ Control room temperature and humidity according to manufacturer's recommendation ○ Perform basic machine & equipment operation ○ Apply appropriate cleaning materials and minor maintenance procedures ○ Practice 5S ○ Practice 3Rs	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate • Hands on	8 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	5.2 Perform major preventive maintenance of equipment /machine	• Lecture and discussion on the following topics: ○ Company policy in relation to relevant technology ○ CTP workflow systems and its operation ○ Relevant equipment and operational processes ○ Preventive maintenance procedures established in the plant ○ Proper waste disposal ○ Awareness on health and safety procedures • Perform the following tasks: ○ Change processing chemicals according to manufacturer's standard maintenance procedure ○ Clean computer to plate (CTP) systems according to manufacturer's maintenance procedure ○ Dispose wastes according to proper waste disposal practice ○ Replace selected parts of equipment / machine and its consumables according to manufacturer's standard maintenance procedure ○ Replace processing chemicals ○ Dispose wastes properly ○ Follow oral & written instructions	• Demonstration • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate • Hands on	16 hours
	5.3 Perform recording of maintenance report	• Lecture and discussion on the following topics: ○ Effective communication ○ Procedures in recording and reporting • Perform the following tasks:	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration	4 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Record inspection of equipment /machine in accordance with company's policies and procedures ○ Record cleaning procedures in accordance with company's policies and procedures ○ Request parts for replacement in accordance with company's policies and procedures ○ Submit reports to relevant personnel for validation in accordance with established procedure ○ Demonstrate written & oral communications ○ Follow work instructions ○ Demonstrate attention to details ○ Demonstrate time conscious		• Direct observation of candidate • Hands on	

3.2 TRAINING DELIVERY

1. The delivery of training shall adhere to the design of the curriculum. Delivery shall be guided by the principles of competency-based TVET.

    a. Course design is based on competency standards set by the industry or recognized industry sector; (Learning system is driven by competencies written to industry standards)

    b. Training delivery is learner-centered and should accommodate individualized and self-paced learning strategies;

    c. Training can be done on an actual workplace setting, simulation of a workplace and/or through adoption of modern technology.

    d. Assessment is based in the collection of evidence of the performance of work to the industry required standards;

    e. Assessment of competency takes the trainee's knowledge and attitude into account but requires evidence of actual performance of the competency as the primary source of evidence.

    f. Training program allows for recognition of prior learning (RPL) or current competencies;

    g. Training completion is based on satisfactory completion of all specified competencies not on the specified nominal duration of learning.

2. The competency-based TVET system recognizes various types of delivery modes, both on-and off-the-job as long as the learning is driven by the competency standards specified by the industry. The following training modalities and their variations/components may be adopted singly or in combination with other modalities when designing and delivering training programs:

    2.1 Institution- Based:

    - Dual Training System (DTS)/Dualized Training Program (DTP) which contain both in-school and in-industry training or fieldwork components. Details can be referred to the Implementing Rules and Regulations of the DTS Law and the TESDA Guidelines on the DTP.

    - Distance learning is a formal education process in which majority of the instruction occurs when the students and instructor are not in the same place. Distance learning may employ correspondence study, audio, video, computer technologies; or other modern technology that can be used to facilitate learning and formal and non-formal training. Specific guidelines on this mode shall be issued by the TESDA Secretariat.

    - Supervised Industry Learning (SIL) or on-the-job training (OJT) is an approach in training designed to enhance the knowledge and skills of the trainee through actual experience in the workplace to acquire specific competencies as prescribed in the training regulations. It is imperative that the deployment of trainees in the workplace is adhered to training programs agreed by the institution and enterprise and status and progress of trainees are closely monitored by the training institutions to prevent opportunity for work exploitation.

- The classroom-based or in-center instruction uses of learner-centered methods as well as laboratory or field-work components.

2.2 Enterprise-Based:

- Formal Apprenticeship – Training within employment involving a contract between an apprentice and an enterprise on an approved apprenticeable occupation.

- Informal Apprenticeship - is based on a training (and working) agreement between an apprentice and a master craftsperson wherein the agreement may be written or oral and the master craftsperson commits to training the apprentice in all the skills relevant to his or her trade over a significant period of time, usually between one and four years, while the apprentice commits to contributing productively to the work of the business. Training is integrated into the production process and apprentices learn by working alongside the experienced craftsperson.

- Enterprise-based Training - where training is implemented within the company in accordance with the requirements of the specific company. Specific guidelines on this mode shall be issued by the TESDA Secretariat.

2.3 Community-Based

- Community-Based – short term programs conducted by non-government organizations (NGOs), LGUs, training centers and other TVET providers which are intended to address the specific needs of a community. Such programs can be conducted in informal settings such as barangay hall, basketball courts, etc. These programs can also be mobile training program (MTP).

3.3 TRAINEE ENTRY REQUIREMENTS

Trainees or students wishing to enroll in this program must possess the following requirements:

- Completed at least grade 10 of basic education or High School graduate or holder of Alternative Learning Systems (ALS) certificate of completion with grade 10 equivalent; and
- Must possess good communication skills

3.4 LIST OF TOOLS, EQUIPMENT AND MATERIALS

Recommended list of tools, equipment and materials for the training of **25 trainees** for Printing Services (Prepress Technical Operations) NC I.

Up-to-date tools, materials, and equipment of equivalent functions can be used as alternatives. This also applies in consideration of community practices and their availability in the local market.

TOOLS

QTY	UNIT	DESCRIPTION/SPECIFICATION
26	pcs	USB Flash Drives, 8G
26	pcs	Steel Rule, 24”
1	pc	Whiteboard (8’ x 4’)
1	unit	Room Temperature and Humidity Meter (Digital)

EQUIPMENT

QTY	UNIT	DESCRIPTION/SPECIFICATION
26	units	Work station computer unit with accessories and installed desktop publishing software Computer Specifications: • Processor: at least i5 or higher (at least 10th generation or higher) • RAM: minimum of 8gb 16gb • Video Card: minimum of 4gb • Storage: minimum of 256gb (SSD) • Monitor: minimum of 24” LED • Compatible Platform • UPS – 650 VA Software Specifications: • Lay –out Designing Software • Vector-Based Software • Image Manipulation and Editing Software • Productivity Software Tools or Word Processing Software
5	units	Flatbed Scanner with OCR Software DPI 1200, A3 size
21	units	LCD Projector
21	units	72” LED Monitor Calibrated
1	units	Monitor Calibrator
1	unit	Computer set as file server • Processor: at least i5 or higher (at least 10th generation or higher)

		• RAM: minimum of 8gb
		• Storage:
		  ○ 1tb as Operating System
		  ○ 1tb 2tb as Main Storage
		  ○ 1tb as Mirrored Disk
		• UPS – 650 VA
		Monitor: 24”

1	unit	Audio System with Lapel Microphone
1	unit	QR Code Scanner
1	unit	Laser Desktop Printer, Standard
1	unit	Inkjet Desktop Printer, Standard
1	unit	Plate Recording Unit, 4 ups 32 diodes with UPS
1	unit	Plate Processor, 4 ups
1	unit	Plate Viewing Table, 4 ups

MATERIALS

QTY	UNIT	DESCRIPTION/SPECIFICATION
26	pcs	Ballpen
26	pcs	Pencil
26	pcs	Hard Case Folder with 3 rings
26	pcs	Fastener
26	pcs	Notebook
1	pc	Log Book
26	pcs	IDs with QR code
5	pcs	1” Camel’s Hairbrush
3	reams	Copy Paper (Letter, A4 & Legal)
3	pcs	Whiteboard Marker (Black, Blue and Red)
1	pc	Whiteboard Eraser
26	pcs	Face Shield
26	pcs	Surgical mask (disposable)
26	pcs	Sponge
26	pcs	Latex Gloves
26	pcs	Plastic Apron
26	pcs	Round Rugs
1	bar	Hand Soap
1	bottle	70% Isopropyl Alcohol (500ml)
1	bottle	Dishwashing Liquid (500 ml)
1	bottle	Wipe n Shine solution
1	unit	Blower
1	unit	Light Duty Vacuum Cleaner

INSTRUCTIONAL MANUALS/FORMS

QTY	UNIT	DESCRIPTION/SPECIFICATION
		Standard Forms:
26	copies	Performance Evaluation Form
26	copies	Variable Data Information Sheet
26	copies	Inspection Schedule Form
26	copies	Inspection Checklist Form
26	copies	Routine Maintenance Report
26	copies	Parts Replacement Report
5	copies	Sample Manuscript
5	copies	Sample Printed Copy for OCR Scanning
5	copies	Printed or Reflective Copies
5	copies	Artwork Materials
		Relevant Policies and Procedure Manuals:
26	copies	Pre-flighting Procedures Manual
26	copies	File Handling Procedure Manual
26	copies	Folder Structure Sample Manual
26	copies	Input Processing Procedure Manual
26	copies	Scanner’s Operating Procedures
26	copies	Inkjet and Laser Printer’s Operating Procedures
26	copies	Line Reproduction Procedures Manual
26	copies	Digital Artwork Tracing Procedures
26	copies	Plate Recorder’s Operating Procedure
26	copies	Plate Processor’s Operating Procedure
		Instructional Materials
26	pcs	Contract Proof Sample
26	sheets	Pre-flighting Checklist
26	sheets	Pre-flighting Report Template
26	sets	Various Printed Samples
26	copies	Proofreading Marks/Symbols Handouts
26	copies	Work Instruction Form
26	copies	Job Order form
26	copies	Task Sheet
26	copies	Job Sheet
26	copies	Theoretical Lecture Handouts
5	pcs	Die-cutting Form Sample

3.5 TRAINING FACILITIES

Based on a class intake of 25 learners/trainees.

Space Requirement	Size in Meters	Area in Sq. Meters	Total Area in Sq. Meters
A. Distance Learning (Laboratory/Workshop/ Activity area)	12 x 15	180	180
B. Practical Work Area	6 x 6	36	36
C. Contextual Learning Area (Lecture Room)	10 x 12	120	120
D. Store/Storage Room	2 x 5	10	10
E. Separate restrooms for female, male and PWD		10	10
F. Circulation Area		22	22
		Total Workshop Area:	**378 sq. m.**

3.6 TRAINER'S QUALIFICATIONS

- Must be a holder of National TVET Trainer Certificate Level I (NTTC I) in Printing Services (Prepress Technical Operations) NC II;
- Must have at least three (3) years prepress technical operations-related industry experience[1] within the last five (5) years **OR** must be a practicing trainer/with teaching experience[2] related to prepress technical operations for two (2) years within the last five (5) years; and
- Must possess good communication skills

[1] A certificate of employment or any document which will serve as evidence that indeed the trainer has industry experience must be submitted.

[2] Refer to the existing provision of INDUSTRY WORK EXPERIENCE REQUIRED (IWER)

3.7 INSTITUTIONAL ASSESSMENT

Institutional assessment is undertaken by trainees to determine their achievement of units of competency. A certificate of achievement is issued for each unit of competency.

SECTION 4 ASSESSMENT AND CERTIFICATION ARRANGEMENT

Competency Assessment is the process of collecting evidence and making judgments whether competency has been achieved. The purpose of assessment is to confirm that an individual can perform to the standards expected at the workplace as expressed in relevant competency standards.

The assessment process is based on evidence or information gathered to prove achievement of competencies. The process may be applied to an employable unit(s) of competency in partial fulfillment of the requirements of the national qualification.

4.1 NATIONAL ASSESSMENT AND CERTIFICATION ARRANGEMENTS

4.1.1 To attain the national qualification of **PRINTING SERVICES (PREPRESS TECHNICAL OPERATIONS) NC I,** the candidate must demonstrate competence in all units listed in Section 1. Successful candidates shall be awarded a National Certificate signed by the TESDA Director General.

4.1.2 Assessment shall cover all competencies with basic and common integrated or assessed concurrently with the core units of competency.

4.1.3 Any of the following are qualified to undergo assessment and certification:

4.1.3.1 Graduates of WTR-registered, NTR-registered programs or formal/non-formal/informal including enterprise-based trainings related to Printing Services (Prepress Technical Operations) NC I; or

4.1.3.2 Experienced workers (wage employed or self-employed) who gained competencies in providing prepress technical operations services for at least two (2) years within the last five (5) years.

4.1.4 **Recognition of Prior Learning (RPL).** Candidates who have gained competencies through education, informal training, previous work or life experiences in prepress technical operations with at least three (3) years within the last five (5) years may apply for recognition in this Qualification through Portfolio Assessment.

Requirements and implementation procedure of Portfolio Assessment must be consistent with TESDA Circular No. 025, Series of 2023 entitled *"Amended Operating Procedure on the Conduct of Portfolio Assessment"*.

4.1.5 The guidelines on assessment and certification are discussed in detail in the "Procedures Manual on Assessment and Certification" and "Guidelines on the Implementation of the "Philippine TVET Competency Assessment and Certification System (PTCACS)".

4.2 COMPETENCY ASSESSMENT REQUISITE

4.2.1 **Self-Assessment Guide**. The self-assessment guide (SAG) is accomplished by the candidate prior to actual competency assessment. SAG is a pre-assessment tool to help the candidate and the assessor determine what evidence is available, where gaps exist, including readiness for assessment.

This document can:

a) Identify the candidate's skills and knowledge
b) Highlight gaps in candidate's skills and knowledge
c) Provide critical guidance to the assessor and candidate on the evidence that need to be presented
d) Assist the candidate to identify key areas in which practice is needed or additional information or skills that should be gained prior to assessment.

4.2.2 **Accredited Assessment Center.** Only Assessment Center accredited by TESDA is authorized to conduct competency assessment. Assessment centers undergo a quality assured procedure for accreditation before they are authorized by TESDA to manage the assessment for National Certification.

4.2.3 **Accredited Competency Assessor**. Only accredited competency assessor is authorized to conduct assessment of competence. Competency assessors undergo a quality assured system of accreditation procedure before they are authorized by TESDA to assess the competencies of candidates for National Certification.

COMPETENCY MAP – SOCIAL AND OTHER COMMUNITY DEVELOPMENT SERVICES SECTOR
PRINTING SERVICES (PREPRESS TECHNICAL OPERATIONS) NC I

BASIC COMPETENCIES

Receive and respond to workplace communication	Work with others	Solve/address routine problems	Enhance self-management skills	Support Innovation	Access and maintain information	Follow occupational safety and health policies and procedures	Apply environmental work standards	Adopt entrepreneurial mindset in the workplace
Participate in workplace communication	Work in team environment	Solve/address general workplace problems	Develop career and life decisions	Contribute to workplace innovation	Present relevant information	Practice occupational safety and health policies and procedures	Exercise efficient and effective sustainable practices in the workplace	Practice entrepreneurial skills in the workplace
Lead workplace communication	Lead small teams	Apply critical thinking and problem-solving techniques in the workplace	Work in a diverse environment	Propose methods of applying learning and innovation in the organization	Use information systematically	Evaluate occupational safety and health work practices	Evaluate environmental work practices	Facilitate entrepreneurial skills for micro-small-medium enterprises (MSMEs)
Utilize specialized communication skill	Develop and lead teams	Perform higher order thinking processes and apply techniques in the workplace	Contribute to the practice of social justice in the workplace	Manage innovative work instructions	Manage and evaluate usage of information	Lead in improvement of Occupational Safety and Health Program, Policies and Procedures	Lead towards improvement of environmental work programs, policies and procedures	Sustain entrepreneurial skills
Manage and sustain effective communication strategies	Manage and sustain high performing teams	Evaluate higher order thinking skills and adjust problem solving techniques	Advocate strategic thinking for global citizenship	Incorporate innovation into work procedures	Develop systems in managing and maintaining information	Manage implementation of occupational safety and health programs in the workplace	Manage implementation of environmental programs in the workplace	Develop and sustain a high-performing enterprise

COMMON COMPETENCIES

Maintain instruments and equipment in work area	Assist in dental laboratory procedures	Assist with administration in dental laboratory practice	Implement and monitor infection control policies and procedures	Respond effectively to difficult/ challenging behavior	Apply basic first aid	Maintain high standard of patient / client services	Apply quality standards	Maintain an effective relationship with customers and clients	Maintain an effective relationship with clients/ customers (marketing)
Update industry knowledge	Manage own performance	Maintain a safe, clean and efficient environment	Perform computer operations	Follow occupational health and safety policies in dental laboratory facilities	Maintain infection control in dental practice	Operate a personal computer	Perform workplace security and safety practices	Update industry knowledge and practice through continuing education	Use pharmaceutical calculation techniques and terminologies
Maintain instruments and equipment in work area	Assist in dental laboratory procedures	Assist with administration in dental laboratory practice	Implement and monitor infection control policies and procedures	Respond effectively to difficult/ challenging behavior	Apply basic first aid	Maintain high standard of patient / client services	Apply quality standards	Maintain a safe, clean and efficient environment	Maintain an effective relationship with clients/ customers (marketing)

CORE COMPETENCIES

Perform facial cleansing	Perform temporary hair removal activity	Perform body scrub	Perform pre and post hair care activities	Perform hair and scalp treatment	Perform basic hair coloring	Perform basic hair bleaching	Perform basic hair perming	Perform hair straightening
Perform basic haircutting	Perform advanced and creative haircutting	Perform advanced and creative hair coloring	Perform advanced and creative hair perming	Perform basic men's haircutting	Perform shave and style beard and mustache	Perform chair manipulative relaxing services	Perform manicure and pedicure	Perform hand and foot spa
Perform preparatory activities	Prepare appropriate products, tools and equipment	Perform nail enhancement technology procedures	Perform post service activities	Perform advanced nail polish procedures	Apply facial make-up	Perform body bleach	Network with stakeholders	Process labor market information
Market and promote services of public employment services office	Perform job matching services	Perform referral and placement services	Perform monitoring of jobseekers	Perform livelihood/training referral and facilitation services	Perform monitoring of referred livelihood and training applicants	Assist in the conduct of career coaching and employment orientation seminar to school-based clients	Provide employment orientation and coaching to walk-in clients	Provide employment orientation and coaching to other clients

Participate in the conduct of research and development of new programs	Implement employment facilitation services and special programs	Organize recruitment activities	Clean living room, dining room, bedroom, bathroom and kitchen	Wash and iron clothes, linen and fabric	Prepare hot and cold meals	Provide food and beverage service	Perform preparatory activities for data collection	Gather data
Perform post-data collection activities	Perform pre-flighting procedures	Perform file handling	Perform input processing procedures	Perform line reproduction and its requirements	Perform preventive maintenance of prepress equipment/machines	Perform halftone reproduction and its requirements	Perform basic color separation process	Perform digital output processing operations
Perform imposition procedures	Perform regular maintenance of condition-based prepress machines/equipment							

GLOSSARY OF TERMS

BITMAP File format	Also known as bitmap image file, device independent bitmap file format and bitmap, is a raster graphics image file format used to store bitmap digital images, independently of the display device, especially on Microsoft Windows and OS/2 operating systems.
Bleed image	Refers to an extra 1/8” (. 125 in) of image or background color that extends beyond the trim area of your printing piece. The project is printed on an oversized sheet that is then cut down to size with the appearance that the image is “bleeding” off the edge of the paper.
Cloud backup	A remote, online, or managed backup service, sometimes marketed as cloud backup or backup-as-a-service, is a service that provides users with a system for the backup, storage, and recovery of computer files.
CMYK	The CMYK acronym stands for Cyan, Magenta, Yellow, and Key: those are the colours used in the printing process. A printing press uses dots of ink to make up the image from these four colours. 'Key' actually means black.
Computer file server	A central server in a computer network that provides file systems or at least parts of a file system to connected clients.
Computer hardware	Includes the physical parts of a computer, such as the case, central processing unit (CPU), monitor, mouse, keyboard, computer data storage, graphics card, sound card, speakers and motherboard. By contrast, software is the set of instructions that can be stored and run by hardware.
Computer network	A group of computers that use a set of common communication protocols over digital interconnections for the purpose of sharing resources located on or provided by the network nodes.
Computer peripherals	External devices that enhance your machine's functionality through better performance, design, sound quality and more. Add-ons range from essential input devices like keyboards, to enjoyable output devices like speakers.
Computer software	Software is a set of programs, which is designed to perform a well-defined function. A program is a sequence of instructions written to solve a particular problem.
Computer-to-Plate (CTP)	An imaging technology used in modern printing processes. In this technology, an image created in a Desktop Publishing (DTP) application is output directly to a printing plate.
Consumables	Goods used by individuals and businesses that must be replaced regularly because they wear out or are used up. They can also be defined as the components of an end product that is used up or permanently altered in the process of manufacturing such as semiconductor wafers and basic chemicals.

Data purging	A process involving methods that permanently erase and remove data from a storage space.
Digital images	Raster images have a finite set of digital values, called picture elements or pixels. The digital image contains a fixed number of rows and columns of pixels.
Digital proof or Proof	A preliminary version of a printed piece, intended to show how the final piece will appear. Proofs are used to view the content, color and design elements before committing the piece to copy plates and press.
Disk mirroring	The replication of logical disk volumes onto separate physical hard disks in real time to ensure continuous availability. It is most commonly used in RAID 1. A mirrored volume is a complete logical representation of separate volume copies.
File format	The structure of a file that tells a program how to display its contents. For example, a Microsoft Word document saved in the .DOC file format is best viewed in Microsoft Word. Even if another program can open the file, it may not have all the features needed to display the document correctly.
Font	A collection of characters with a similar design. These characters include lowercase and uppercase letters, numbers, punctuation marks, and symbols. Changing the font can alter the look and feel of a block of text.
Hard copy	A printed document.
High key image	A style of image that uses unusually bright lighting to reduce or completely blow out dark shadows in the image. High key shots usually lack dark tones and the high key look is generally thought of as positive and upbeat.
Image capture	An application program from Apple that enables users to upload pictures from digital cameras or scanners which are either connected directly to the computer or the network. It provides no organizational tools like iPhoto but is useful for collating pictures from a variety of sources with no need for drivers.
Image resolution	The number of pixels in an image. Resolution is sometimes identified by the width and height of the image as well as the total number of pixels in the image.
Image trapping	The compensation for misregistration between printing units on a multicolor press. This misregistration causes gaps or white space on the final printed packaging.
Kerning	The process of adjusting the spacing between characters in a proportional font, usually to achieve a visually pleasing result. Kerning adjusts the space between individual letterforms, while tracking adjusts spacing uniformly over a range of characters.
Leading	The distance between each line of text. It is pronounced ledding (like "sledding" without the "s"). The name comes from a time when typesetting was done by hand and pieces of lead were used to separate the lines.

Low key image	A style of image that utilizes predominantly dark tones to create a dramatic looking image. Where high key lighting seeks to over light the subject to the point of reduced contrast, low key lighting intensifies the contrast in an image through intensely reduced lighting.
Malicious software	Any malicious program that causes harm to a computer system or network. Malicious Malware Software attacks a computer or network in the form of viruses, worms, trojans, spyware, adware or rootkits.
Optical Character Recognition (OCR)	The electronic or mechanical conversion of images of typed, handwritten or printed text into machine-encoded text, whether from a scanned document, a photo of a document, a scene-photo (for example the text on signs and billboards in a landscape photo)
Overprinting	The process of printing one color on top of another in reprographics. This is closely linked to the reprographic technique of 'trapping'. It is also used to create a rich black (often regarded as a color that is "blacker than black") by printing black over another dark color.
Page layout	The arrangement of visual elements on a page. It generally involves organizational principles of composition to achieve specific communication objectives.
Paragraph styles	Generally applied to one or more paragraphs. Note that border and background attributes are set separately but they are actually a subset of the paragraph attributes.
Pixelated	The pixel (a word invented from "picture element") is the basic unit of programmable color on a computer display or in a computer image. Think of it as a logical - rather than a physical - unit. The physical size of a pixel depends on how you've set the resolution for the display screen.
Pre-flighting	The process of checking if the digital data required to print a job are all present and valid. Nowadays PDF files are sent to printing companies. The PDF file format is a solid standard to exchange pages, ranging from single ads to complete publications.
Prepress	The term used in the printing and publishing industries for the processes and procedures that occur between the creation of a print layout and the final printing.
Press	Printing presses use ink to transfer text and images to paper. Medieval presses used a handle to turn a wooden screw and push against paper laid over the type and mounted on a platen. Metal presses, developed late in the 18th century, and used steam to drive a cylinder press.
Process colors	A way of mixing inks to create colors during the actual printing process itself. A process color is printed using a combination of the four standard process inks: cyan, magenta, yellow and black (CMYK). Also used in offset printing, process colors are the more common method of printing.

Raster image	A dot matrix data structure that represents a generally rectangular grid of pixels (points of color), viewable via a computer display, paper, or other display medium.
Raster image processor	A component used in a printing system which produces a raster image also known as a bitmap. Such a bitmap is used by a later stage of the printing system to produce the printed output.
Raw file	Collection of unprocessed data. This means the file has not been altered, compressed, or manipulated in any way by the computer.
Reproducible copies	To make a copy, representation, or imitation of; duplicate
Image resolution	The detail an image holds. The term applies to raster digital images, film images, and other types of images. Higher resolution means more image detail. Image resolution can be measured in various ways. Resolution quantifies how close lines can be to each other and still be visibly resolved.
Scanning procedures	A process of recording an exact image of a document
Soft copy	A document saved on a computer. It is the electronic version of a document, which can be opened and edited using a software program.
Spot color	Any color generated by an ink (pure or mixed) that is printed using a single run, whereas a process color is produced by printing a series of dots of different colors.
Storage devices	Any type of hardware that stores data. The most common type of storage device, which nearly all computers have, is a hard drive.
Text styles	A resource that specifies text attributes, including the font, size, line spacing, font style, text alignment, and text and background colors.
Typesetting/Encoding	The process of putting text onto a page so it's print-ready.
Variable data	Aata that is acquired through measurements, such as length, time, diameter, strength, weight, temperature, density, thickness, pressure, and height.
Vector graphics	Vector graphics are computer graphics images that are defined in terms of points which are connected by lines and curves to form polygons and other shapes.
Word processing software	A word processor is a device or computer program that provides for input, editing, formatting, and output of text, often with some additional features.

ACRONYMS

AI	Adobe Illustrator file extension
CDR	CorelDRAW file extension
CMS	Color Management System
CMYK	Cyan, Magenta, Yellow, Black
CtP	Computer-to-Plate
DVI	Digital Visual Interface
EPS	Encapsulated PostScript file extension
GCR	Gray Component Replacement
HDMI	High-Definition Multimedia Interface
INDD	Adobe InDesign file extension
JO	Job Order
JPEG	Joint Photographic Experts Group
LED	Light-Emitting Diode
OCR	Optical Character Recognition
OHS	Occupational Health and Safety
PDF	Portable Document Format
PS	Postscript
PSD	Photoshop Document file extension
RAM	Random-Access Memory
RGB	Red, Green, and Blue
RIP	Raster Image Processor
TIFF/TIF	Tagged Image File Format file extension
UCR	Under Color Removal
USB	Universal Serial Bus
VGA	Video Graphics Array

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